

Master's Thesis

Circle Bundles with Positive Scalar Curvature over Enlargeable Manifolds

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Introduction

The scalar curvature of a Riemannian manifold (M, g) is a smooth function $\text{scal}_g: M \rightarrow \mathbb{R}$. Roughly speaking, the value $\text{scal}_g(x)$ at a point $x \in M$ measures how much the volume of a small geodesic ball around x deviates from that of a Euclidean ball of the same radius. While scalar curvature is the weakest of the standard curvature conditions for a Riemannian manifold, the presence of *positive* scalar curvature can still have strong topological implications. The archetypal example is the celebrated Gauß-Bonnet Theorem:

Theorem (Gauß-Bonnet). *Let (M, g) be a closed Riemannian 2-manifold. Then*

$$\int_M \text{scal}_g \, dA = 4\pi\chi(M),$$

where $\chi(M)$ is the Euler characteristic of M .

So any closed surface carrying a metric of positive scalar curvature must be diffeomorphic to either the sphere S^2 or $\mathbb{R}P^2$. For non-compact complete manifolds, positive scalar curvature alone may be too weak a condition for such rigidity results though. The problem is that scal can still tend to zero quite rapidly at infinity. However, if the rate of decay at infinity can be controlled, positive scalar curvature still has profound consequences. For example, Chen has recently shown a decomposition result for 3-manifolds under the assumption that a metric of positive scalar curvature with controlled decay at infinity exists [Che25]. This naturally raises the question of how fast the scalar curvature of a given complete Riemannian manifold X decays at infinity. One result in this direction is the following theorem by Gromov [Gro18, p. 15f.].

Theorem (Enlargeable Decay Theorem). *Let (X, g) be a complete Riemannian manifold and $V \rightarrow M$ a 2-dimensional real vector bundle over an enlargeable manifold M . Assume that*

- (i) *There exists a proper map $f: X \rightarrow V$, $\deg(f) \neq 0$,*
- (ii) *The total space of the circle bundle corresponding to V is enlargeable.*

Then

$$\min_{B_R(x_0)} \text{scal}_g \leq \frac{8\pi^2}{(R - R_0)^2}$$

for some $R_0 = R_0(X, x_0)$ and all $R > R_0$, where $B_R(x_0)$ is the ball of radius R around a point $x_0 \in M$.

Condition (ii) is always fulfilled if the bundle $V \rightarrow M$ is trivial (compare Lemma 3.3 (i)), which led Gromov to pose the following question, here written using the notation introduced above.

Question ([Gro18, p. 16]). “What happens to nontrivial bundles $V \rightarrow M$? [...] In general, the examples in [BH10] indicate a possibility of non-enlargeable circle bundles over enlargeable M ; yet, it seems hard(er) to find such examples, where the corresponding V would admit complete metrics with $\text{scal} \geq \sigma > 0$.”

Note that the existence of such a bundle $V \rightarrow M$ would provide counterexamples to the above decay theorem if condition (ii) is dropped entirely.

As anticipated by Gromov, examples of non-enlargeable circle bundles over enlargeable manifolds do in fact exist. Bundles with this behaviour have recently been constructed in a paper by Kumar and Sen [KS25] by utilizing tools from symplectic geometry. A very rough sketch of the construction can be given as follows. Take M to be a so-called Donaldson divisor of a suitable closed symplectic manifold, e.g., the torus T^{2n} , $n \geq 3$. Such a Donaldson divisor is a certain codimension 2 symplectic submanifold $M^{2n-2} \subseteq T^{2n}$, which in some cases can be shown to be enlargeable. Now define a circle bundle $E := \partial\nu(M)$ over M as the boundary of a tubular neighbourhood $\nu(M) \subseteq T^{2n}$ of M . The authors then proceed to show that E admits a metric of positive scalar curvature by the following argument. Consider the nullbordism $W := T^{2n} - \nu(M)$ obtained by removing the tubular neighbourhood from the ambient manifold. Now explicitly construct a Morse function $\psi: W \rightarrow \mathbb{R}$ with critical points of index at most n , which shows that E is obtained from the empty set by surgeries of codimension at least $n \geq 3$. Thus, E admits a metric of positive scalar curvature by the Surgery Theorem (see Theorem 2.7) and so in particular it is not enlargeable (see Theorem 3.2).

In this thesis, we take a different approach and provide a much more direct construction of such bundles drawing only on general topological concepts, without invoking symplectic methods. This approach, however, is limited to the totally non-spin case. We then apply our results to obtain examples of 2-dimensional vector bundles over enlargeable manifolds, where the total space admits a metric of uniformly positive scalar curvature, thereby providing an affirmative answer to Gromov's question. Note that

The thesis is organized as follows. Chapter 1 serves as a review of principal bundles, classifying spaces and characteristic classes and lays the foundation for the subsequent chapters. In Chapter 2, we briefly introduce the reader to the theory of positive scalar curvature and prove two existence theorems, which are later applied in the main construction. Chapter 3 provides a brief discussion of the concept of enlargeability. With the necessary background established, we then present the main results discussed above in Chapter 4. Finally, Chapter 5 discusses some further questions regarding the preceding construction.

1. Preliminaries

This chapter develops the necessary preliminaries for what follows by recalling the notions of principal bundles, classifying spaces, and characteristic classes. We assume that the reader is familiar with the concept of a fiber bundle with structure group. A classical reference for this is, for example, [Ste51]. Moreover, we presuppose some basic familiarity with homotopy and homology theory. The first section, on principal bundles, draws mainly on [DK01] and [Mit11]. In the second section, on classifying spaces, we follow [Ebe12]. Finally, the third section, on characteristic classes, is based primarily on [Hat03], with additional material drawn from [LM89] and [Fre26].

1.1. Principal Bundles and Associated Bundles

Definition 1.1. Let G be a topological group. A G -principal bundle is a fiber bundle with fiber G and structure group G acting by left translations.

The total space of a G -principal bundle $P \xrightarrow{\pi} B$ carries a natural right action of G defined as follows. Around $y \in P$, choose a local trivialisation

$$\phi: U \times G \rightarrow \pi^{-1}(U)$$

and let $\phi^{-1}(y) = (\pi(y), h)$. Then set

$$y \cdot g := \phi(\pi(y), hg).$$

This is well defined since, by definition, G acts on itself by *left* translations under a change of local trivialisation. A morphism of principal bundles should respect this right action, thus we define:

Definition 1.2. A morphism from a G -principal bundle $P \rightarrow B$ to an H -principal bundle $P' \rightarrow B'$ is a triple (F, f, ρ) of continuous maps $F: P \rightarrow P'$ and $f: B \rightarrow B'$ together with a homomorphism $\rho: G \rightarrow H$, such that

$$\begin{array}{ccc} P & \xrightarrow{F} & P' \\ \downarrow & & \downarrow \\ B & \xrightarrow{f} & B' \end{array}$$

commutes and such that F is ρ -equivariant, that is $F(x \cdot g) = F(x) \cdot \rho(g)$ for all $x \in P$ and $g \in G$. If $G = H$, we assume ρ to be the identity, and if $B = B'$, we assume f to be the identity.

The condition of equivariance in the above definition is actually quite strong, as illustrated by the following proposition.

Proposition 1.3. *Any morphism of G -principal bundles over the same base space is an isomorphism.*

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Proof. First suppose we are given a morphism

$$F: B \times G \rightarrow B \times G$$

of trivial bundles. Then by definition F is of the form

$$F(x, g) = (x, \phi_x(g))$$

for a continuous family of maps $\phi_x: G \rightarrow G$. By equivariance, $\phi_x(g) = \phi_x(e)g$, and thus an inverse of F is given by

$$F^{-1}(x, g) := (x, \phi_x(e)^{-1}g).$$

Since every G -principal bundle is locally trivial, this shows that we can always find local inverses in the general case. As inverses are unique, they fit together to define a global inverse. $\square$

To any fiber bundle with structure group G , one can canonically associate a G -principal bundle by a general construction that allows one to change the fiber of a given bundle. We now recall this construction of changing the fiber. Let $p: E \rightarrow B$ be a fiber bundle with fiber F and structure group G and let F' be another space upon which G acts effectively. Then we can construct a new bundle as follows. Choose an open cover $(U_\alpha)_{\alpha \in A}$ of B together with local trivialisations $\phi_\alpha: U_\alpha \times F \rightarrow p^{-1}(U_\alpha)$. Let $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow G$ be the corresponding transition functions defined by

$$(\phi_\alpha^{-1} \circ \phi_\beta)(x, f) = (x, \tau_{\alpha\beta}(x) \cdot f).$$

Define

$$E' := \left(\coprod_{\alpha \in A} U_\alpha \times F' \right) / \sim$$

where the equivalence relation $\sim$ is defined by

$$(b, f', \beta) \sim (b, \tau_{\alpha\beta}(b) \cdot f', \alpha)$$

and let $p': E' \rightarrow B$, $[(b, f', \alpha)] \mapsto b$. Then the maps

$$\psi_\alpha: U_\alpha \times F' \rightarrow (p')^{-1}(U_\alpha), (b, f') \mapsto [(b, f', \alpha)].$$

define local trivialisations of p' with the same transition functions $\tau_{\alpha\beta}$, giving $p': E' \rightarrow B$ the structure of a fiber bundle with fiber F' and structure group G . We say that this new bundle is obtained from $E \rightarrow B$ by *changing the fiber* from F to F' (with the G -action on F' being implicit). Note that we can also obtain $E \rightarrow B$ from $E' \rightarrow B$ by changing the fiber from F' back to F . Thus no information is lost in this process. Considering the special case $F' = G$ with G acting on itself by left translations, we call the bundle obtained from $E \rightarrow B$ by changing the fiber from F to G the *associated principal bundle*. Morally speaking, if we want to understand the bundle $E \rightarrow B$, it is sufficient to understand its associated principal bundle, since this contains the same information, as we have seen above. The benefit of studying principal bundles is that they are highly symmetric objects, and thus usually easier to understand. For example, many constructions simplify for principal bundles, such as the concept of universal bundles, as we will see in the next section. Another such instance is the construction for changing the fiber from above. In the case of principal bundles this can be carried out more directly and has the benefit of also allowing for non-effective group actions. This is known as the *Borel Construction*.

Lemma 1.4. *Let $p: P \rightarrow B$ be a G -principal bundle and F a space upon which G acts from the left. Then*

$$E := (P \times F)/\sim, \quad (y, f) \sim (y \cdot g, g^{-1}f)$$

together with the map $\pi: E \rightarrow B$ given by $[(y, f)] \mapsto p(y)$ is a fiber bundle with fiber F and structure group G . Moreover, if the action of G on F is effective, this bundle is isomorphic to the bundle obtained from $P \rightarrow B$ by changing the fiber from G to F using the construction outlined above.

Proof. To see that $P \times_G F$ is locally trivial, let $\psi: U \times G \rightarrow P$ be a local trivialisation. Then the homeomorphism

$$\psi \times \text{id}_F: U \times G \times F \rightarrow p^{-1}(U) \times F$$

descends to a homeomorphism

$$\tilde{\kappa}: (U \times G \times F)/\sim \rightarrow (p^{-1}(U) \times F)/\sim = \pi^{-1}(U),$$

where the equivalence relation on $U \times G \times F$ is given by $(x, g, f) \sim (x, gh, h^{-1}f)$. Moreover, the map

$$U \times G \times F \rightarrow U \times F, \quad (x, g, f) \mapsto (x, gf)$$

descends to a homeomorphism $(U \times G \times F)/\sim \rightarrow U \times F$ with inverse given by $(x, f) \mapsto [(x, e, f)]$. Under this homeomorphism, $\tilde{\kappa}$ corresponds to a homeomorphism $\kappa: U \times F \rightarrow \pi^{-1}(U)$, which is explicitly given by $(x, f) \mapsto [(\psi(x, e), f)]$. Since

$$\begin{array}{ccc} U \times F & \xrightarrow{\kappa} & \pi^{-1}(U) \\ & \searrow \text{proj.} & \downarrow \pi \\ & & U \end{array}$$

commutes, this shows that $\pi: E \rightarrow B$ is locally trivial. To see that, for effective G -actions, this construction agrees with the change-of-fiber construction introduced above, we compute the transition functions of the local trivialisations just defined. Let $U_\alpha, U_\beta \subseteq B$ be two trivialising neighbourhoods for $P \rightarrow B$ with local trivialisations ψ_α, ψ_β and transition function $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow G$. Moreover, let $\kappa_\alpha, \kappa_\beta$ be the corresponding local trivialisations for $E \rightarrow B$ as constructed above. Then we compute

$$(\kappa_\alpha^{-1} \circ \kappa_\beta)(x, f) = \kappa_\alpha^{-1}([(\psi_\beta(x, e), f)]) = (x, \tau_{\alpha\beta}(x)f).$$

Thus the statement follows from the fact that fiber bundles, where the structure group acts effectively, are uniquely determined by the transition functions of local trivialisations. $\square$

We denote the bundle $E \rightarrow B$ obtained from the above construction by $P \times_G F \rightarrow B$. The Borel construction is natural with respect to pullbacks, as stated in the following lemma.

Lemma 1.5. *Let $p: P \rightarrow B$ be a G -principal bundle, F a space upon which G acts from the left and $f: B' \rightarrow B$ a continuous map. Then*

$$(f^*P) \times_G F \cong f^*(P \times_G F).$$

Proof. An explicit model for the pullback bundle is

$$f^*P := \{(y, b') \in P \times B' \mid p(y) = f(b')\}.$$

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Then

$$(f^*P) \times_G F = \{(y, b', f) \in P \times B' \times F \mid p(y) = f(b')\} / \sim,$$

for $(y, b', f) \sim (y \cdot g, b', g^{-1} \cdot f)$. Moreover,

$$\begin{aligned} f^*(P \times_G F) &= \{([y, f], b') \in P \times_G F \times B' \mid p(y) = f(b')\} \\ &\cong \{(y, f, b') \in P \times F \times B' \mid p(y) = f(b')\} / \sim \end{aligned}$$

for $(y, f, b') \sim (y \cdot g, g^{-1} \cdot f, b')$. Clearly these are isomorphic bundles. $\square$

Remark 1.6. We remark that the action of G on F in the above lemma need not be effective. If we restrict ourselves to effective actions, a more general naturality result holds for the change-of-fiber construction of arbitrary bundles. Namely, first pulling a bundle back under some continuous map and then changing the fiber amounts to the same as first changing the fiber and then pulling back. Comparing the transition functions of the local trivialisations in both cases shows that they in fact agree.

Let us look at some concrete examples for the theory developed above.

Example 1.7. (i) Let $\pi: E \rightarrow B$ be a real rank n vector bundle over a paracompact base space B . Choosing a bundle metric gives $E \rightarrow B$ the structure of a fiber bundle with fiber $\mathbb{R}^n$ and structure group $O(n)$. The associated principal bundle is explicitly given by the *orthonormal frame bundle* $O(E)$ of E , whose fiber over a point $x \in B$ is the space of all ordered orthonormal bases of E_x . The topology on $O(E)$ is the subspace topology $O(E) \subseteq E \times \dots \times E$. Local trivialisations of the canonical projection map $p: O(E) \rightarrow B$ can be defined as follows. Choose an orthonormal local frame $e_1, \dots, e_n$ of E defined over an open set $U \subseteq B$. Then a trivialisation $\phi: U \times O(n) \rightarrow p^{-1}(U)$ is given by

$$(x, A) \mapsto \left(\sum_{i=1}^n a_{i1} e_i(x), \dots, \sum_{i=1}^n a_{in} e_i(x) \right),$$

where $A = (a_{ij})$. So $\phi(x, A)$ is the basis of $E_x = \pi^{-1}(x)$, whose coordinates in the frame (e_i) are precisely the columns of the matrix A . The transition functions of these trivialisations agree with the transition functions of the trivialisations of E . Since the transition functions uniquely determine the bundle, we conclude that $O(E)$ is in fact isomorphic to the associated principal bundle as defined at the beginning of this section. Analogously, if $E \rightarrow B$ is a complex rank n vector bundle, choosing a hermitian metric on E allows us to reduce the structure group to $U(n)$, and the corresponding associated principal bundle is the *unitary frame bundle* $U(E)$.

- (ii) Let $p: \tau_1^n \rightarrow \mathbb{C}P^n$ be the tautological complex line bundle equipped with the canonical hermitian metric. Since this bundle is of rank 1, its unitary frame bundle is simply the subspace $U(\tau_1^n) \subseteq \tau_1^n$ of all vectors of norm 1, together with the restriction of p to this subspace. Recall that

$$\tau_1^n = \{(L, z) \in \mathbb{C}P^n \times \mathbb{C}^{n+1} \mid z \in L\}.$$

So, explicitly,

$$U(\tau_1^n) = \{(L, z) \in \tau_1^n \mid |z| = 1\}.$$

Consider the diffeomorphism

$$U(\tau_1^n) \rightarrow S^{2n+1}, (L, z) \mapsto z$$

with inverse $S^{2n+1} \rightarrow U(\tau_1^n)$, $z \mapsto (\pi(z), z)$, where $\pi: S^{2n+1} \rightarrow \mathbb{C}P^n$ is the canonical projection. Under this diffeomorphism the restriction of p to $U(\tau_1^n)$ corresponds to π . Hence π defines a $U(1) \cong S^1$ -principal bundle, which agrees with the associated principal bundle of $\tau_1^n \rightarrow \mathbb{C}P^n$.

- (iii) Let $E \rightarrow B$ be an S^1 -principal bundle over a smooth manifold B without boundary. We can change the fiber of E from S^1 to D^2 to obtain another bundle $DE \rightarrow B$, where S^1 acts on D^2 by rotations. By definition of the change-of-fiber construction, trivialisations $U \times D^2 \rightarrow DE$ have the same transition functions as the corresponding trivialisations of $E \rightarrow B$. Moreover, they restrict to trivialisations $U \times S^1 \rightarrow \partial DE$ of the boundary. Since the action of S^1 on D^2 restricts to the action of S^1 on itself, we see that $E \cong \partial DE$. So, in particular, the total space of any S^1 -principal bundle is null-bordant. We will make use of this fact in Chapter 4.

1.2. Universal Bundles and Classifying Spaces

A natural question is which fiber bundles with specified fiber and structure group can exist over a given base space. As before, this problem turns out to be easier for principal bundles. We assume all base spaces to be CW-complexes in this section.

Definition 1.8. A G -principal bundle $E \rightarrow B$ is called *universal* if E is weakly contractible as a topological space.

The significance of this definition is established by the following theorem. We say that two morphisms of G -principal bundles are G -homotopic if they are homotopic through morphisms of G -principal bundles.

Theorem 1.9. *Let $E \rightarrow B$ be a universal G -principal bundle. Then for any other G -principal bundle $P \rightarrow X$ there exists a morphism $P \rightarrow E$ of G -principal bundles, which is unique up to G -homotopy.*

Before showing the theorem, let us discuss its implications. Firstly, we can equivalently formulate the theorem by saying that $E \rightarrow B$ is a terminal object in the homotopy category of G -principal bundles over CW-complexes and thus, in particular, uniquely determined (up to G -homotopy equivalence). In view of this, one usually writes $EG := E$ and $BG := B$, and refers to $EG \rightarrow BG$ as *the* universal G -principal bundle. The base space BG is called the *classifying space* of G . Since any bundle morphism $(F, f): P \rightarrow EG$ induces a morphism $P \rightarrow f^*EG$ (and vice versa, which is true for arbitrary fiber bundles), we conclude from Proposition 1.3 that

$$P \cong f^*EG.$$

We call f a *classifying map* for P . By homotopy invariance of the pullback, we get a well-defined bijection

$$\phi_X: [X, BG] \rightarrow \mathcal{P}_G(X), [f] \mapsto [f^*EG],$$

where $\mathcal{P}_G(X)$ denotes the set of isomorphism classes of G -principal bundles over X . The existence result in Theorem 1.9 shows that ϕ_X is surjective and the uniqueness result shows that it is injective. Moreover, ϕ_X is clearly natural in X . Thus we have shown:

Corollary 1.10. *The maps ϕ_X define a natural isomorphism of contravariant functors*

$$[-, BG], \mathcal{P}_G(-): \mathbf{HCW} \rightarrow \mathbf{Set},$$

where $\mathbf{HCW}$ denotes the homotopy category of CW-complexes.

Theorem 1.9 will be a simple application of the following theorem.

Theorem 1.11. *Let $P \rightarrow X$ be a G -principal bundle and $A \subseteq X$ a subcomplex. Then any morphism $P|_A \rightarrow EG$ of G -principal bundles extends to a morphism $P \rightarrow EG$ of G -principal bundles.*

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The proof of this theorem is based on an inductive, cell-by-cell approach. We refer the reader to [Ebe12, Theorem 2.7.1].

Proof of Theorem 1.9. Applying Theorem 1.11 to $A = \emptyset$ shows the existence of a morphism $P \rightarrow EG$. Now suppose we are given morphisms $F_0, F_1: P \rightarrow EG$. Consider the G -principal bundle

$$\pi \times \text{id}: P \times [0, 1] \rightarrow X \times [0, 1],$$

where $\pi: P \rightarrow X$ is the bundle projection (this is just the pullback bundle of P under the projection $X \times [0, 1] \rightarrow X$). Then F_0 and F_1 define a morphism of G -principal bundles

$$(F_0, F_1): P \times \{0, 1\} = (P \times [0, 1])|_{X \times \{0, 1\}} \rightarrow EG,$$

which by Theorem 1.11 can be extended to $P \times [0, 1]$, thus yielding a G -homotopy between F_0 and F_1 . $\square$

Of course, for this theory to be of any use, we need such universal bundles to exist. The following general existence theorem was first shown by Milnor [Mil56].

Theorem 1.12. *For any topological group G there exists a universal G -principal bundle $EG \rightarrow BG$, where BG is a CW-complex.*

A priori, the space BG constructed in [Mil56] may not be a CW-complex. However, if we choose a CW-approximation $f: X \rightarrow BG$, the total space of the bundle $f^*EG \rightarrow X$ is still weakly contractible. This follows by applying the five lemma to the following diagram.

$$\begin{array}{ccccccccc} \pi_{k+1}(X) & \longrightarrow & \pi_k(G) & \longrightarrow & \pi_k(f^*EG) & \longrightarrow & \pi_k(X) & \longrightarrow & \pi_{k-1}(G) \\ \downarrow \cong & & \downarrow \text{id} & & \downarrow & & \downarrow \cong & & \downarrow \text{id} \\ \pi_{k+1}(BG) & \longrightarrow & \pi_k(G) & \longrightarrow & \pi_k(EG) & \longrightarrow & \pi_k(BG) & \longrightarrow & \pi_{k-1}(G) \end{array}$$

We can now use these classification results for principal bundles to obtain analogous results for general fiber bundles by appealing to the change-of-fiber construction discussed in the preceding section.

Corollary 1.13. *Let G be a topological group acting effectively on a space F . Then for any CW-complex X the map*

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), \quad [f] \mapsto [f^*(EG \times_G F)]$$

is a natural bijection, where $\mathcal{F}_{F,G}(X)$ denotes the set of isomorphism classes of fiber bundles with fiber F and structure group G over X .

Proof. By Corollary 1.10, the map

$$[X, BG] \rightarrow \mathcal{P}_G(X), \quad [f] \mapsto [f^*EG]$$

is a natural bijection. Moreover, since G acts effectively the map

$$\mathcal{P}_G(X) \rightarrow \mathcal{F}_{F,G}(X), \quad [P] \mapsto [P \times_G F]$$

is a bijection, which is natural by Lemma 1.5. Composing both thus yields the natural bijection

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), \quad [f] \mapsto [(f^*EG) \times_G F] = [(f^*EG \times_G F)],$$

once more applying Lemma 1.5 in the last equality. $\square$

We conclude this section by discussing some examples.

Example 1.14.

- (i) From the long exact sequence of homotopy groups

$$\pi_k(EG) \longrightarrow \pi_k(BG) \longrightarrow \pi_{k-1}(G) \longrightarrow \pi_{k-1}(EG)$$

we conclude that $\pi_k(BG) \cong \pi_{k-1}(G)$ for $k \geq 1$. Moreover, BG is connected as the image of the connected space EG under $EG \rightarrow BG$.

- (ii) The universal cover $\mathbb{R} \rightarrow S^1$ is a $\mathbb{Z}$ -principal bundle, $\mathbb{Z}$ acting on itself by addition. Since $\mathbb{R}$ is contractible we obtain $B\mathbb{Z} \cong S^1$ and $E\mathbb{Z} \cong \mathbb{R}$.
- (iii) The universal $O(k)$ -principal bundle $EO(k) \rightarrow BO(k)$ also has an explicit description. Consider the Stiefel manifold $V_k^n \subseteq \mathbb{R}^{n \times k}$, which is defined as the compact space of orthonormal k -tuples of vectors in $\mathbb{R}^n$. Furthermore, let Gr_k^n be the Grassmannian of k -planes in $\mathbb{R}^n$, i.e. the set of k -dimensional linear subspaces of $\mathbb{R}^n$ with the topology induced by the projection map

$$p: V_k^n \rightarrow \text{Gr}_k^n, (v_1, \dots, v_k) \mapsto \text{span}(v_1, \dots, v_k).$$

Identifying a linear subspace $V \in \text{Gr}_k^n$ with $\mathbb{R}^k$ yields an identification of the fiber $p^{-1}(V)$ with $O(k)$. Moreover, the identifications $V \cong \mathbb{R}^k$ can be chosen consistently in a small neighbourhood around V , providing local trivialisations for p . The transition function of two such trivialisations takes values in $O(k)$, hence p defines an $O(k)$ -principal bundle. The canonical inclusion $\mathbb{R}^n \rightarrow \mathbb{R}^{n+1}$ induces inclusions $V_k^n \rightarrow V_k^{n+1}$ and $\text{Gr}_k^n \rightarrow \text{Gr}_k^{n+1}$. Consider the colimits over these inclusions

$$V_k^\infty := \text{colim}_n (V_k^n), \quad \text{Gr}_k^\infty := \text{colim}_n (\text{Gr}_k^n)$$

and the induced map $V_k^\infty \rightarrow \text{Gr}_k^\infty$. One can choose trivialisations of $V_k^n \rightarrow \text{Gr}_k^n$ for varying n , which fit together to induce a trivialisation in the colimit, such that $V_k^\infty \rightarrow \text{Gr}_k^\infty$ defines an $O(k)$ -principal bundle. We claim that V_k^∞ is weakly contractible. Consider the map

$$V_k^n \rightarrow S^{n-1}, (v_1, \dots, v_k) \mapsto v_1.$$

Similar arguments as above show that this map is locally trivial with fiber V_{k-1}^{n-1} . Hence we can consider the long exact sequence of homotopy groups

$$\pi_{i+1}(S^{n-1}) \longrightarrow \pi_i(V_{k-1}^{n-1}) \longrightarrow \pi_i(V_k^n) \longrightarrow \pi_i(S^{n-1})$$

So $\pi_i(V_k^n) \cong \pi_i(V_{k-1}^{n-1})$ for $i \leq n-3$. Applying this inductively yields

$$\pi_i(V_k^n) \cong \pi_i(V_1^{n-(k-1)}) = \pi_i(S^{n-k+1}) = 0$$

for $i \leq n-k-1$. Now from compactness of S^i and the fact that V_k^n is closed in V_k^{n+1} , we conclude that the image of any map $S^i \rightarrow V_k^\infty$ is contained in some finite stage V_k^n . Therefore $\pi_i(V_k^\infty) = 0$ for all $i \geq 0$, such that the universal $O(k)$ -principal bundle is explicitly given by $V_k^\infty \rightarrow \text{Gr}_k^\infty$.

Similarly we get an explicit description of the universal $SO(k)$ -principal bundle. Let $\tilde{\text{Gr}}_k^n$ be the set of tuples consisting of a k -dimensional linear subspace of $\mathbb{R}^n$ together with a chosen orientation. The topology on $\tilde{\text{Gr}}_k^n$ is induced by

$$p: V_k^n \rightarrow \tilde{\text{Gr}}_k^n, (v_1, \dots, v_k) \mapsto (\text{span}(v_1, \dots, v_k), \text{o}_{(v_1, \dots, v_k)}),$$

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where $o_{(v_1, \dots, v_k)}$ is the unique orientation on $\text{span}(v_1, \dots, v_k)$, such that $(v_1, \dots, v_k)$ is positively oriented. Arguments analogous to the above show that p defines an $SO(k)$ -principal bundle. Taking the colimit, we obtain the universal $SO(k)$ -principal bundle $V_k^\infty \rightarrow \tilde{\text{Gr}}_k^\infty$.

- (iv) Any real k -dimensional vector bundle $E \rightarrow X$ can be considered as a fiber bundle with fiber $\mathbb{R}^k$ and structure group $O(k)$. So Corollary 1.13 and the preceding example show that every such E is the pullback of the universal real k -dimensional vector bundle $V_k^\infty \times_{O(k)} \mathbb{R}^k$ under some map $X \rightarrow \text{Gr}_k^\infty$. We show that this universal vector bundle also has a more explicit description, namely as the tautological bundle $\gamma_k \rightarrow \text{Gr}_k^\infty$. Recall that this bundle is defined as

$$\gamma_k := \{(X, v) \in \text{Gr}_k^\infty \times \mathbb{R}^\infty \mid v \in X\}$$

together with the canonical projection map to Gr_k^∞ . An element in V_k^∞ is a k -tuple of orthonormal vectors in some $\mathbb{R}^n$, which we will subsequently regard as the columns of an $n \times k$ matrix. Then the map

$$V_k^\infty \times \mathbb{R}^k \rightarrow \gamma_k, \quad (A, v) \mapsto (\text{im}(A), Av)$$

descends to the quotient, defining a vector bundle map $V_k^\infty \times_{O(k)} \mathbb{R}^k \rightarrow \gamma_k$. This map is clearly a linear isomorphism on fibers, hence an isomorphism of vector bundles.

Similarly, oriented real k -dimensional vector bundles are classified by maps to $\tilde{\text{Gr}}_k^\infty$. We claim that the universal oriented real k -dimensional vector bundle is given by the tautological bundle $\gamma_k^{SO} \rightarrow \tilde{\text{Gr}}_k^\infty$ defined by

$$\gamma_k^{SO} := \{(X, o_X, v) \in \tilde{\text{Gr}}_k^\infty \times \mathbb{R}^k \mid v \in X\}$$

together with the canonical projection map to $\tilde{\text{Gr}}_k^\infty$. To this end consider the map

$$V_k^\infty \times \mathbb{R}^k \rightarrow \gamma_k^{SO}, \quad (A, v) \mapsto (\text{im}(A), o_A, Av),$$

where o_A is the unique orientation on $\text{im}(A)$, such that $A: \mathbb{R}^k \rightarrow \text{im}(A)$ is orientation preserving. Again this map descends to the quotient, defining an isomorphism $V_k^\infty \times_{SO(k)} \mathbb{R}^k \rightarrow \gamma_k^{SO}$.

- (v) Completely analogous to the real case, we can also define the complex Stiefel manifolds $V_k^\infty(\mathbb{C})$ and complex Grassmannians $\text{Gr}_k^\infty(\mathbb{C})$ and show that the universal $U(k)$ -principal bundle is given by $V_k^\infty(\mathbb{C}) \rightarrow \text{Gr}_k^\infty(\mathbb{C})$. The universal complex k -dimensional vector bundle is the tautological bundle $\tau_k \rightarrow \text{Gr}_k^\infty(\mathbb{C})$.
- (vi) From the special cases $k = 1$ of the preceding examples we obtain

$$B\mathbb{Z}_2 \cong \mathbb{R}P^\infty, \quad BS^1 \cong \mathbb{C}P^\infty.$$

1.3. Stiefel-Whitney and Chern-Classes

Again we assume all base spaces to be CW-complexes in this section. In the preceding chapter we discussed the question of how bundles can be classified. Another fundamental question is, of course, how to determine whether two given bundles are isomorphic. As usual in topology, a partial solution to this problem is provided by constructing computable invariants. For vector bundles the most important invariants are so-called *characteristic classes*. The general definition of a characteristic class is as follows. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and write $\text{Vect}_{\mathbb{K}}(X)$ for the set of isomorphism classes of $\mathbb{K}$ -vector bundles over a space X .

Together with the pullback operation, $\text{Vect}_{\mathbb{K}}(-)$ defines a contravariant functor $\text{CW} \rightarrow \text{Set}$. A characteristic class c for $\mathbb{K}$ -vector bundles with values in a cohomology theory H^* is a natural transformation from the functor $\text{Vect}_{\mathbb{K}}(-)$ to the functor H^* , regarded as a functor with values in Set . In more concrete terms, c assigns to each $\mathbb{K}$ -vector bundle $E \rightarrow X$ a cohomology class $c(E) \in H^*(X)$, depending only on the isomorphism class of E , such that for any continuous map $f: X' \rightarrow X$ the following naturality condition holds:

$$c(f^*E) = f^*c(E).$$

Using the theory of classifying spaces developed above, it follows that for real (resp. complex) k -dimensional vector bundles, any characteristic class c is uniquely determined by its value on the universal bundle γ_k (resp. τ_k) discussed in Example 1.14 (iv) (resp. ??).

For real vector bundles, the most important characteristic classes are the *Stiefel-Whitney classes*, and for complex vector bundles, the *Chern classes*. Both are characterized by a few axioms.

Definition 1.15 (Stiefel-Whitney Classes). There is a unique sequence of characteristic classes $w_1, w_2, \dots$ for real vector bundles with values in $H^*(-; \mathbb{Z}_2)$, such that

- (i) $w_i(E) \in H^i(X; \mathbb{Z}_2)$ for any real vector bundle $E \rightarrow X$,
- (ii) $w_i(E) = 0$ for $i > \text{rank}(E)$,
- (iii) $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$, where $w = 1 + w_1 + w_2 + \dots$,
- (iv) For the universal real line bundle $\gamma_1 \rightarrow \mathbb{R}P^\infty$, $w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2) \cong \mathbb{Z}_2$.

Note that by property (ii), the sum $w = 1 + w_1 + w_2 + \dots$ is always finite. The class $w(E)$ is called the *total Stiefel-Whitney class* of E . We sometimes write $w_0(E) := 1 \in H^0(X; \mathbb{Z}_2)$ to simplify notation.

Definition 1.16 (Chern Classes). There is a unique sequence of characteristic classes $c_1, c_2, \dots$ for complex vector bundles with values in $H^*(-; \mathbb{Z})$, such that

- (i) $c_i(E) \in H^{2i}(X; \mathbb{Z})$ for any complex vector bundle $E \rightarrow X$,
- (ii) $c_i(E) = 0$ for $i > \text{rank}(E)$,
- (iii) $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$, where $c = 1 + c_1 + c_2 + \dots$,
- (iv) For the universal complex line bundle $\tau_1 \rightarrow \mathbb{C}P^\infty$, $c_1(\tau_1)$ is a generator of $H^2(\mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z}$ specified in advance.

Again the *total Chern class* $c = 1 + c_1 + c_2 + \dots$ is well-defined and we write $c_0(E) := 1 \in H^0(X; \mathbb{Z})$. Note that in both definitions, property (iv) uniquely determines the values of the respective characteristic classes on line bundles and, together with property (iii), prescribes their values on bundles that split as sums of line bundles. The values on arbitrary vector bundles can then be deduced from a general splitting principle [Hat03, Proposition 3.3]. We refer the reader to [Hat03] for a detailed proof of existence and uniqueness.

Not only are the definitions of Stiefel-Whitney and Chern classes formally quite similar, but there is also a direct connection between the two.

Proposition 1.17 ([Hat03, Proposition 3.8]). *Regarding a complex rank n vector bundle $E \rightarrow X$ as a real rank $2n$ vector bundle, $w_{2i+1}(E) = 0$ and $w_{2i}(E)$ is the image of $c_i(E)$ under the coefficient homomorphism $H^{2i}(X; \mathbb{Z}) \rightarrow H^{2i}(X; \mathbb{Z}_2)$.*

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Let us look at a few examples.

Example 1.18.

- (i) Let $\iota: \mathbb{R}P^n \rightarrow \mathbb{R}P^\infty$ be the canonical inclusion. Then the pullback $\iota^*\gamma_1$ is isomorphic to the tautological line bundle $\gamma_1^n \rightarrow \mathbb{R}P^n$. Since ι induces an isomorphism on cohomology, we conclude from naturality that $w_1(\gamma_1^n) = \iota^*w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^n; \mathbb{Z}_2) \cong \mathbb{Z}_2$. Similarly, for the tautological complex line bundle $\tau_1^n \rightarrow \mathbb{C}P^n$, the class $c_1(\tau_1^n)$ is a generator of $H^2(\mathbb{C}P^n; \mathbb{Z}) \cong \mathbb{Z}$.
- (ii) Let $E \rightarrow X$ be a trivial real vector bundle. Then $E \cong f^*E$ for a constant map $f: X \rightarrow X$, and thus $w_i(E) = f^*w_i(E) = 0$ for all $i \geq 1$. Similarly, all Chern classes for complex trivial vector bundles vanish.
- (iii) Let $TS^n \rightarrow S^n$ be the tangent bundle of the sphere and let $\nu_n \rightarrow S^n$ be the normal bundle with respect to the standard embedding $S^n \rightarrow \mathbb{R}^{n+1}$. Since ν_n and $TS^n \oplus \nu_n$ are trivial, we conclude from the previous example and property (iii) in Definition 1.15 that

$$1 = w(TS^n \oplus \nu_n) = w(TS^n) \smile w(\nu_n) = w(TS^n) \smile 1 = w(TS^n).$$

Thus $w_i(TS^n) = 0$ for all $i \geq 1$.

The last example shows that although all Stiefel-Whitney classes of a vector bundle vanish, it need not be trivial. However, for real (resp. complex) vector bundles of rank 1, it turns out that the first Stiefel-Whitney (resp. Chern) class already determines the bundle:

Proposition 1.19 ([Hat03, Proposition 3.10]). *The maps*

$$w_1: \text{Vect}_{\mathbb{R}}(X) \rightarrow H^1(X; \mathbb{Z}_2)$$

and

$$c_1: \text{Vect}_{\mathbb{C}}(X) \rightarrow H^2(X; \mathbb{Z})$$

are bijections.

While the first Stiefel-Whitney class of higher dimensional bundles does not identify their isomorphism type, it does characterize another important invariant, namely orientability:

Proposition 1.20 ([Hat03, Proposition 3.11]). *A real vector bundle $E \rightarrow X$ is orientable if and only if $w_1(E) = 0$.*

Similarly, vanishing of the second Stiefel-Whitney class corresponds to the existence of another structure, namely that of a *spin structure*.

Definition 1.21. (i) For $n \geq 2$ the *spin group* $\text{Spin}(n)$ is defined as the total space of the non-trivial two-sheeted cover $\rho: \text{Spin}(n) \rightarrow \text{SO}(n)$. If $n \geq 3$, this is the universal cover, and in the case $n = 2$ we have $\text{Spin}(2) \cong \text{SO}(2)$. For $n = 1$ we define $\text{Spin}(1) := \mathbb{Z}_2$, the total space of the two-sheeted cover $\rho: \mathbb{Z}_2 \rightarrow \text{SO}(1)$.

- (ii) Let $E \rightarrow X$ be an oriented real vector bundle equipped with a fiber metric and let $\text{SO}(E)$ be its oriented orthonormal frame bundle. A *spin structure* on E is a $\text{Spin}(n)$ -principal bundle $\text{Spin}(E) \rightarrow X$ together with a map

$$\phi: \text{Spin}(E) \rightarrow \text{SO}(E)$$

of principal bundles, which is ρ -equivariant (compare Definition 1.2).

- (iii) An orientable manifold M is called *spin* if its tangent bundle TM admits a spin structure and is called *totally non-spin* if the universal cover of M is not spin.

Remark 1.22. When $n = 1$, $SO(E) \cong X$ and a spin structure is simply a two-sheeted cover of X .

Note that the existence of a spin structure does not depend on the chosen orientation or fiber metric on E , since different choices yield isomorphic oriented orthonormal frame bundles. In fact, the only obstruction to the existence of a spin structure is the second Stiefel-Whitney class, as mentioned above:

Theorem 1.23 ([LM89, Chapter 2, Theorem 1.7]). *An orientable real vector bundle $E \rightarrow X$ admits a spin structure if and only if $w_2(E) = 0$.*

Example 1.24. (i) It follows from the above theorem and Example 1.18 (iii) that all spheres S^n , $n \geq 1$, are spin.

- (ii) The total Chern class of the tangent bundle $T\mathbb{C}P^n$ of $\mathbb{C}P^n$ can be computed to be

$$c(T\mathbb{C}P^n) = (1 + \alpha)^{n+1}$$

for a suitably chosen generator $\alpha \in H^2(\mathbb{C}P^n; \mathbb{Z})$; see [MS74, Theorem 14.10]. In particular, $c_1(T\mathbb{C}P^n) = (n+1)\alpha$. From Proposition 1.17 we see that $w_2(T\mathbb{C}P^n) = 0$ if and only if n is odd. Therefore, $\mathbb{C}P^n$ is spin if and only if n is odd.

Remark 1.25. Let $E \rightarrow X$ be a real vector bundle. If E is orientable, we have seen that this bundle is classified by a map $X \rightarrow BSO(n)$. Similarly, if a spin structure exists, E is classified by a map $X \rightarrow BSpin(n)$, which can be seen as follows. Define a vector bundle

$$\gamma_n^{\text{Spin}} := E\text{Spin}(n) \times_{\text{Spin}(n)} \mathbb{R}^n$$

using Lemma 1.4, where $\text{Spin}(n)$ acts on $\mathbb{R}^n$ by virtue of the covering map $\rho: \text{Spin}(n) \rightarrow SO(n)$ and the canonical action of $SO(n)$. Let $\phi: \text{Spin}(E) \rightarrow SO(E)$ be a spin structure on E and let $f: X \rightarrow B\text{Spin}(n)$ be a classifying map for $\text{Spin}(E)$. From Lemma 1.5 we obtain

$$f^* \gamma_n^{\text{Spin}} \cong \text{Spin}(E) \times_{\text{Spin}(n)} \mathbb{R}^n.$$

Moreover, the spin structure ϕ induces an isomorphism $\text{Spin}(E) \times_{\text{Spin}(n)} \mathbb{R}^n \rightarrow SO(E) \times_{SO(n)} \mathbb{R}^n \cong E$ of vector bundles by

$$[(x, v)] \mapsto [(\phi(x), v)].$$

Since this map restricts to a linear bijection on each fiber, it is an isomorphism of vector bundles.

Recall that a manifold M is orientable if and only if its tangent bundle TM restricted to any embedded loop $S^1 \subseteq M$ is trivial. Or, in other words, “ TM does not contain any Moebius bands”. A similar characterization can be given for spin manifolds, which is why the property of being spin is sometimes referred to as a higher dimensional analogue of orientability.

Proposition 1.26. *Let M be a connected oriented manifold of dimension at least 5.*

- (i) *M is spin if and only if for every embedded surface $S \subseteq M$ the restriction $TM|_S$ of the tangent bundle to S is trivial.*
- (ii) *M is totally non-spin if and only if there exists an embedding $\iota: S^2 \rightarrow M$, such that $w_2(\iota^*TM) \neq 0$. In particular, the restriction ι^*TM of the tangent bundle to this embedded sphere is non-trivial.*

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Proof. Since we will not be making use of the first characterization, we shall only prove part (ii) of the proposition and refer the reader to [Fre26, Proposition 2.6] for a proof of part (i). The proof closely follows [Fre26]. Let $\tilde{M}$ be the universal cover of M . Suppose that $T\tilde{M}$ does not admit a spin structure. Then we conclude from Theorem 1.23 that $w_2(T\tilde{M}) \neq 0$. So, by the universal coefficient theorem, there exists a homology class $a \in H_2(\tilde{M}; \mathbb{Z}_2)$ with

$$\langle w_2(T\tilde{M}), a \rangle \neq 0.$$

Once more applying the universal coefficient theorem as well as the Hurewicz theorem, we obtain an isomorphism

$$\pi_2(\tilde{M}) \otimes \mathbb{Z}_2 \rightarrow H_2(\tilde{M}; \mathbb{Z}) \otimes \mathbb{Z}_2 \cong H_2(\tilde{M}; \mathbb{Z}_2).$$

Explicitly, for some class $\alpha \in \pi_2(\tilde{M})$ represented by a map $f: S^2 \rightarrow \tilde{M}$, the above isomorphism maps $\alpha \otimes 1$ to $f_*[S^2]_{\mathbb{Z}_2}$, where $[S^2]_{\mathbb{Z}_2}$ is the $\mathbb{Z}_2$ -fundamental class of S^2 . Therefore we can write $a = f_*[S^2]_{\mathbb{Z}_2}$ for some $f: S^2 \rightarrow \tilde{M}$. Let $p: \tilde{M} \rightarrow M$ be the projection and define $\iota := p \circ f: S^2 \rightarrow M$. Since $\dim(M) \geq 5$, we may assume ι to be an embedding. We compute

$$\langle w_2(\iota^*TM), [S^2]_{\mathbb{Z}_2} \rangle = \langle f^*w_2(p^*TM), [S^2]_{\mathbb{Z}_2} \rangle = \langle w_2(T\tilde{M}), f_*[S^2]_{\mathbb{Z}_2} \rangle = \langle w_2(T\tilde{M}), a \rangle \neq 0,$$

where we used $p^*TM \cong T\tilde{M}$ and naturality of the Kronecker pairing in the second equation. Therefore $w_2(\iota^*TM) \neq 0$. Conversely, if there exists some embedding $\iota: S^2 \rightarrow M$ with $w_2(\iota^*TM) \neq 0$, we can lift ι along p to an embedding $\tilde{\iota}: S^2 \rightarrow \tilde{M}$. Again using $p^*TM \cong T\tilde{M}$ we obtain

$$w_2(\tilde{\iota}^*T\tilde{M}) = w_2(\iota^*TM) \neq 0.$$

Therefore $w_2(T\tilde{M}) \neq 0$ by naturality of w_2 . $\square$

Finally, we record another scenario in which the vanishing of all Stiefel-Whitney classes of a vector bundle is sufficient for triviality.

Lemma 1.27. *Let $E \rightarrow S$ be a real vector bundle of rank at least 3 over a two-dimensional CW-complex S . Then E is trivial if and only if $w_1(E) = w_2(E) = 0$.*

Proof. Combining Theorem 1.23 and Remark 1.25, we conclude that there exists a classifying map $f: S \rightarrow B\text{Spin}(n)$ for E , where n is the rank of E . From Example 1.14 (i) it follows that $\pi_k(B\text{Spin}(n)) = 0$ for $k \leq 2$. Since S is two-dimensional, the map f is homotopic to a constant by [Hat02, Lemma 4.6], therefore $E \cong f^*\xi_{\text{Spin}}$ is trivial. $\square$

2. Positive Scalar Curvature

The purpose of this chapter is twofold. The first section provides a brief introduction to positive scalar curvature, following [Wal17] and [Car21]. In the second section, we prove two concrete existence results for metrics of positive scalar curvature, which will be used in the construction presented in Chapter 4.

2.1. A Brief Introduction

Let M^n be an n -dimensional manifold. Given any Riemannian metric g on M , we can associate to it a smooth function $\text{scal}_g: M \rightarrow \mathbb{R}$ called the *scalar curvature* of M . In the case where M is an embedded surface in $\mathbb{R}^3$, scal_g equals twice the classical Gaussian curvature $K: M \rightarrow \mathbb{R}$. So in this case, the scalar curvature has a clear geometric interpretation, illustrated below.

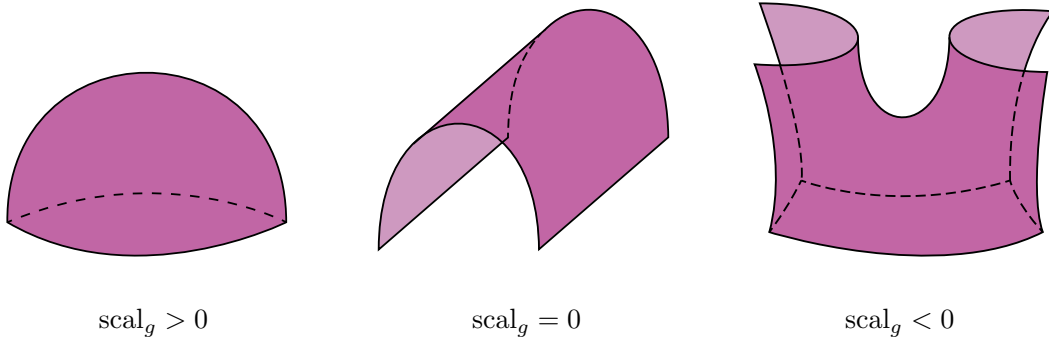


Figure 2.1.: Scalar curvature in dimension 2.

In dimension 2, scalar curvature is the only intrinsic notion of curvature. However, in dimensions $n \geq 3$, there exists a whole hierarchy of different curvatures, the most prominent of which are sectional curvature, Ricci curvature and scalar curvature. The sectional curvature is the strongest notion and determines the entire geometry of M . In this thesis, we are concerned with the scalar curvature, which is the weakest among the three. In most textbooks on Riemannian geometry, it is defined by taking twice the trace of the Riemannian curvature tensor. While this definition is best suited for explicit computations, it provides no further intuition on what scalar curvature is. Since we have no need for doing such explicit computations, we give a more geometric definition, which essentially extends our intuition in the 2-dimensional case to arbitrary dimensions.

Definition 2.1. Let (M^n, g) be an n -dimensional Riemannian manifold. The scalar curvature $\text{scal}_g(x) \in \mathbb{R}$ at the point $x \in M$ is defined as the value appearing in the third coefficient of the Taylor expansion

$$\frac{\text{vol}(B_r(x))}{\text{vol}(B_r^{\text{eucl.}}(0))} = 1 - \frac{\text{scal}_g(x)}{6(n+2)}r^2 + \dots,$$

where $\text{vol}(B_r(x))$ denotes the volume of a geodesic ball in M of radius r centered at x , and $\text{vol}(B_r^{\text{eucl.}}(0))$ denotes the volume of a ball of the same radius r in standard Euclidean space $\mathbb{R}^n$.

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So, intuitively, $\text{scal}_g(x) > 0$ means that the volume of a ball in M around x of sufficiently small radius r is smaller than that of a Euclidean ball of the same radius. Similarly, if $\text{scal}_g(x) < 0$, the volume of the ball in M is larger, and if $\text{scal}_g(x) = 0$, the volumes are (asymptotically) identical. Note that this is in accordance with Figure 2.1.

Example 2.2. (i) We immediately obtain from the definition that the scalar curvature of $\mathbb{R}^n$ with the usual Euclidean metric is zero.

(ii) If (M, g) is one-dimensional, the volume of a geodesic ball of radius r is simply $2r$. Therefore, $\text{scal}_g = 0$.

(iii) The scalar curvature of the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ can be computed to equal $n(n-1)$.

(iv) Let (M_1, g_1) and (M_2, g_2) be Riemannian manifolds. Then, for the scalar curvature of the product $(M_1 \times M_2, g_1 + g_2)$ equipped with the product metric $g_1 + g_2$, one obtains $\text{scal}_{g_1+g_2}(x_1, x_2) = \text{scal}_{g_1}(x_1) + \text{scal}_{g_2}(x_2)$ for $x_i \in M_i$. We refer the reader to any classical textbook on Riemannian geometry for a proof of these last two examples.

Broadly speaking, the study of scalar curvature falls into two categories: *rigidity* results and *flexibility* results. As the title of this chapter suggests, one is primarily concerned with results about *positive* scalar curvature. The reason for this is the following theorem. See, for example, [SJ98].

Theorem 2.3. *Let M^n be a closed connected manifold of dimension $n \geq 3$, and let $f: M \rightarrow \mathbb{R}$ be a smooth function.*

(i) *If there exists a point $p \in M$ with $f(p) < 0$, then M admits a metric g with scalar curvature $\text{scal}_g = f$.*

(ii) *If M admits a metric whose scalar curvature is non-negative but not identically zero, then M admits another metric g with scalar curvature $\text{scal}_g = f$.*

Part (i) of the theorem asserts that the only scalar curvature condition from which we can expect any kind of obstruction results is non-negativity. Moreover, aside from the scalar-flat case (where the scalar curvature is identically zero), part (ii) shows that studying non-negative scalar curvature yields nothing new beyond what is already captured by positive scalar curvature. In the following, we refer to a metric g on M whose scalar curvature is everywhere positive as a **PSC-metric**.

The reason why the study of positive scalar curvature has attracted particular attention compared to other curvature conditions, such as positive Ricci or sectional curvature, is that it strikes an exceptionally good balance between the rigidity and flexibility aspects. While positive Ricci or sectional curvature has substantial topological implications, this also makes it significantly more difficult to construct metrics with these curvature conditions. For example, the Bonnet-Myers theorem states that a positive lower bound on the Ricci curvature of a complete Riemannian manifold M already implies that M must be compact with finite fundamental group. For PSC-metrics, however, there are existence results under quite general assumptions, which we will discuss below. First, let us consider some examples illustrating the rigidity aspects of positive scalar curvature. The most prominent obstruction to the existence of a PSC-metric comes from spin geometry. Let (M^n, g) be a closed Riemannian spin manifold with n even. Then the spin structure and the metric g determine a Hermitian vector bundle $S \rightarrow M$ together with a compatible connection ∇^S and a self-adjoint first-order differential operator $D: \Gamma(S) \rightarrow \Gamma(S)$ acting

on sections of S . Moreover, there exists a canonical splitting $S \cong S^+ \oplus S^-$ with respect to which D splits as

$$D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}.$$

We call $S \rightarrow M$ the *spinor bundle* of M and D the *Dirac operator*. One of the central motivations behind constructing this operator was to find a first-order differential operator squaring to the Laplacian. The spinor bundle S carries additional structure making it possible to define such an operator. However, its square agrees with the Laplacian only up to an error term determined by the curvature of M . More precisely, the *Schrödinger–Lichnerowicz formula* holds:

$$D^2 = \Delta^S + \frac{1}{4} \text{scal}_g,$$

where $\Delta^S: \Gamma(S) \rightarrow \Gamma(S)$ is the connection Laplacian determined by ∇^S . Given any vector bundle $E \rightarrow N$ over a closed Riemannian manifold N carrying a fiber metric and a compatible connection, it is a general fact that the induced connection Laplacian is a non-negative operator with respect to the L^2 -inner product $\langle \cdot, \cdot \rangle$ on $\Gamma(E)$ induced by the fiber metric and the volume form of N . Applying this to S , let $\psi \in \Gamma(S)$. Then

$$0 \leq \langle D\psi, D\psi \rangle = \langle D^2\psi, \psi \rangle = \langle \Delta^S\psi, \psi \rangle + \frac{1}{4} \int_M \text{scal}_g \|\psi\|^2,$$

where we used self-adjointness of D in the first equation and the Schrödinger–Lichnerowicz formula in the second. If $\text{scal}_g > 0$ and $\psi \neq 0$, the right-hand side is strictly positive, since Δ^S is a non-negative operator. But the right-hand side equals $\langle D\psi, D\psi \rangle = \|D\psi\|^2$, so this forces $D\psi \neq 0$. Hence the kernel of D must be trivial. In particular, the integer

$$\text{ind}(D) := \dim(\ker D^+) - \dim(\ker D^-),$$

known as the *index* of D , must vanish (one can show that the kernels of $D^\pm$ are finite-dimensional, so this is well defined). The major insight is that this index can also be computed in terms of a purely topological invariant of M , namely the so-called $\hat{A}$ -class $\hat{A}(TM) \in H^{4*}(M; \mathbb{Q})$ of the tangent bundle $TM \rightarrow M$, which is a certain characteristic class. The connection between the index of the Dirac operator and the $\hat{A}$ -class of the tangent bundle is established by the famous *Atiyah–Singer index theorem*, which states:

$$\text{ind}(D) = \langle \hat{A}(TM), [M] \rangle,$$

where the right-hand side is the evaluation of the cohomology class $\hat{A}(TM)$ against the fundamental class $[M]$ of M (for the components of $\hat{A}(TM)$ in degree $\neq n$, this is 0, so in particular $\langle \hat{A}(TM), [M] \rangle = 0$ if n is not divisible by 4). The number $\hat{A}(M) := \langle \hat{A}(TM), [M] \rangle \in \mathbb{Q}$ is also referred to as the *$\hat{A}$ -genus* of M . In summary:

Theorem 2.4 (Lichnerowicz, 1963). *Let (M^{4k}, g) be a closed Riemannian spin manifold with positive scalar curvature. Then $\hat{A}(M) = 0$.*

We refer the reader to [LM89] for an in-depth discussion of the above results. A further, more accessible and more geometric example for a rigidity result concerning positive scalar curvature is the following classical theorem by Llarull [Lla98].

Theorem 2.5 (Llarull, 1998). *Let M^n be a closed connected Riemannian spin manifold of dimension $n \geq 2$. Suppose $\text{scal}_g \geq n(n-1)$ and let $f: M \rightarrow S^n$ be a smooth 1-Lipschitz map with $\deg(f) \neq 0$. Then f is an isometry.*

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As we noted in Example 2.2 (iii), the constant $n(n-1)$ is equal to the scalar curvature of the n -sphere S^n with the standard round metric g_0 . Llarull then applied this theorem in the special case $M = S^n$ and $f = \text{id}$ to show that the only metric g on S^n with $\text{scal}_g \geq n(n-1)$ and $g \geq g_0$ is g_0 itself. Said differently, one cannot increase the curvature of the sphere without contracting distances. We conclude our discussion of scalar curvature rigidity results with another strongly geometric statement, which is an example for a so-called *band width estimate* [Gro18].

Theorem 2.6 (Gromov, 2018). *Let $M = T^{n-1} \times [-1, 1]$ and let g be a Riemannian metric on M with scalar curvature bounded from below by a positive constant $\sigma > 0$. Then the distance between the two boundary components $\partial_- M = T^{n-1} \times \{-1\}$ and $\partial_+ M = T^{n-1} \times \{1\}$ satisfies*

$$\text{dist}_g(\partial_- M, \partial_+ M) \leq 2\pi \sqrt{\frac{n-1}{n\sigma}}.$$

Intuitively, we think of M as a cylinder (a ‘band’) over T^{n-1} with a left boundary component $\partial_- M$ and a right boundary component $\partial_+ M$. Then this theorem essentially states that the boundaries of such a band, which is highly positively curved, must be close together. There are various generalisations of Theorem 2.6 to allow for bands over other manifolds [Zei22], [CZ24]. Gromov even conjectures that T^{n-1} can be replaced by any manifold of dimension $n-1 \geq 5$ not admitting a PSC-metric [Gro18, Conjecture C].

Let us now turn to the flexibility aspect of positive scalar curvature. The primary method for constructing PSC-metrics is the following theorem, known as the Surgery Theorem, which has been established independently by Schoen and Yau [SY79, Cor. 6] as well as Gromov and Lawson [GL80a, Thm. A]. We refer the reader to the Appendix for a brief introduction to surgery and cobordism theory.

Theorem 2.7 (Schoen–Yau, 1979, Gromov–Lawson, 1980). *Let M be a closed manifold which admits a PSC-metric. Then any manifold M' obtained from M by surgery of codimension at least 3 also admits a PSC-metric.*

Remark 2.8. The construction in [GL80a] of the PSC-metric on M' only changes the given PSC-metric on M locally near the embedding along which the surgery is performed. Therefore, the same construction also applies to show the following version of the Surgery Theorem. Let M be a manifold with boundary and g a PSC-metric on M . Suppose that M' is obtained from M by surgery of codimension at least 3 along an embedding into the interior of M . Then M' admits a PSC-metric g' which agrees with g near the boundary $\partial M' \cong \partial M$. See also [EF21, Thm. 1.6].

In order to apply this theorem, we need a way to control the dimension of the surgeries performed, or equivalently, the indices of handles in a given cobordism. This question is completely independent of scalar curvature. It has, for example, been studied quite thoroughly in the context of the h -cobordism theorem. One central tool to control the indices of the handles in a cobordism is the following theorem due to Wall [Wal71, Thm. 3].

Theorem 2.9 (Wall, 1971). *Let W be a cobordism from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W$ is r -connected for $r \leq \dim(W) - 4$. Then W contains no handles of index $\geq \dim(W) - r$.*

Combining the above theorem with the Surgery Theorem 2.7, we can make the following observation. Suppose that $\dim(W) \geq 6$. If M_0 admits a PSC-metric and the inclusion $M_1 \rightarrow W$ is 2-connected, then M_1 also admits a PSC-metric. In fact, using the conclusion of Theorem 2.9, it is possible to obtain an even stronger result, originally due to Gaja [Gaj87]. See also [Fre26, Prop. B.1]. To state the result we introduce the following

terminology. A metric g on a manifold W with boundary is of *product form near the boundary* if there is some collar neighbourhood $\iota: \partial W \times [0, 1) \rightarrow W$, such that $\iota^*g = g|_{\partial W} + dt^2$, where $g|_{\partial W}$ is the induced metric on ∂W .

Theorem 2.10. *Let W be a cobordism from M_0 to M_1 of dimension $\dim(W) \geq 6$, such that the inclusion $M_1 \rightarrow W$ is 2-connected. Then any PSC-metric on M_0 can be extended to a PSC-metric on W , which is of product form near the boundary.*

In view of the above discussion, one strategy to construct a PSC-metric on a given closed manifold M_1 of dimension at least 5 is to first find a cobordism W from some PSC-manifold M_0 to M_1 . One can then attempt to modify this cobordism by surgeries to obtain another cobordism W' from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W'$ is 2-connected. Of course, such a modification by surgeries may not always be possible, so our next goal is to explain certain scenarios in which it is.

2.2. Existence Results for PSC-Metrics

In this section, we prove the following two existence results for PSC-metrics, one in the spin case and one in the totally non-spin case.

Theorem 2.11. *Let W^{n+1} be a connected spin cobordism from M_0 to M_1 , $n \geq 5$. Suppose M_1 is connected, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 , and M_0 admits a PSC-metric. Then M_1 also admits a PSC-metric. In fact, any PSC-metric on M_0 can be extended to a PSC-metric on W , which is of product form near the boundary.*

Theorem 2.12. *Let W^{n+1} be a connected oriented cobordism from M_0 to M_1 , $n \geq 5$. Suppose M_1 is connected and totally non-spin, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 , and M_0 admits a PSC-metric. Then M_1 also admits a PSC-metric. In fact, any PSC-metric on M_0 can be extended to a PSC-metric on W , which is of product form near the boundary.*

The similarity between the two statements is no coincidence. In fact, both are a consequence of a more general result, see [EF21, Theorem 1.5]. The proofs presented here essentially adapt the proof in [EF21] to the special cases at hand. Let us first prove Theorem 2.12.

Proof of Theorem 2.12. The crux of the proof is to find generators of $\pi_2(W, M_1)$ lifting to elements in $\pi_2(W)$ that are represented by embeddings with trivial normal bundle along which we can then perform surgery (compare Proposition A.2). Ideally, we would like $\pi_2(W, M_1)$ to be a finitely generated group for this purpose, which is, however, not the case in general. Nevertheless, viewed as a $\mathbb{Z}[\pi_1(M_1)]$ -module, $\pi_2(W, M_1)$ is always finitely generated. We have included a detailed proof of this fact in Proposition A.5. Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_2(W, M_1)$ be such a set of generators. We will now construct lifts in $\pi_2(W)$ with the desired properties. Let

$$F: W \rightarrow B := BSO(n+1)$$

be a classifying map for the tangent bundle TW and set $f := F \circ i: M_1 \rightarrow B$, where $i: M_1 \rightarrow W$ is the inclusion. Then f is a classifying map for $TM_1 \oplus \mathbb{R}$ with $\mathbb{R}$ the trivial line bundle. We claim that f is surjective on π_2 . From Example 1.14 (i) we conclude $\pi_2(B) \cong \pi_1(SO(n+1)) \cong \mathbb{Z}_2$. Hence it suffices to show that f is non-zero on π_2 . By Proposition 1.26 (ii), there exists an embedding $\iota: S^2 \rightarrow M_1$ such that $w_2(\iota^*TM_1) \neq 0$. We compute

$$w_2((f \circ \iota)^* \gamma_{n+1}^{\text{SO}}) = w_2(\iota^*(TM_1 \oplus \mathbb{R})) = w_2(\iota^*TM_1) \neq 0.$$

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Therefore, $f \circ \iota$ represents a non-trivial element in $\pi_2(B)$, which shows the claim. Now consider the following diagram, where the row is exact, using the fact that $\pi_1(M_1) \rightarrow \pi_1(W)$ is an isomorphism.

$$\begin{array}{ccccccc} \pi_2(M_1) & \xrightarrow{i_*} & \pi_2(W) & \longrightarrow & \pi_2(W, M_1) & \longrightarrow & 0 \\ & \searrow f_* & \downarrow F_* & & & & \\ & & \pi_2(B) & & & & \end{array}$$

Using surjectivity of f_* , a diagram chase shows that there exist lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of the generators $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_2(W, M_1)$, which lie in the kernel of F_* . Moreover, since $n \geq 5$, the classes α_i can be represented by embeddings

$$\phi_i: S^2 \rightarrow W.$$

We show that the normal bundle ν_i of ϕ_i is trivial. Since W is oriented, it follows from Proposition 1.20 that $w_1(\phi_i^*TW) = 0$. Furthermore, $F \circ \phi_i$ is null-homotopic by construction, so that

$$w_2(\phi_i^*TW) = w_2((F \circ \phi_i)^* \gamma_{n+1}^{\text{SO}}) = 0.$$

Using $w_k(TS^2) = 0$ (see Example 1.18 (iii)), we compute

$$w_k(\nu_i) = w_k(\nu_i \oplus TS^2) = w_k(\phi_i^*TW) = 0$$

for $k = 1, 2$. Thus, Lemma 1.27 shows that ν_i is indeed trivial. We can therefore perform surgery to obtain a new cobordism W' from M_0 to M_1 with

$$\pi_2(W', M_1) = 0$$

and $\pi_1(W', M_1) \cong \pi_1(W, M_1) = 0$; see Proposition A.2 (ii) and Remark A.3. In other words, the inclusion $M_1 \rightarrow W'$ is 2-connected. So, by Theorem 2.10, there exists a PSC-metric g on W' , which is of product form near the boundary. To construct a PSC-metric on the original cobordism W , observe that, since W' was obtained from W by surgeries of dimension 2, W can be obtained from W' by surgeries of codimension 3. Therefore, the Surgery Theorem applies to construct a PSC-metric on W and by Remark 2.8 this metric is still in product form near the boundary. $\square$

Proof of Theorem 2.11. As before, $\pi_2(W, M_1)$ is finitely generated as a $\mathbb{Z}[\pi_1(M_1)]$ -module. Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ be such generators. One could argue as in the proof of Theorem 2.12, replacing $BSO(n+1)$ by $B\text{Spin}(n+1)$ and observing that the corresponding f is trivially surjective on π_2 , since $\pi_2(B\text{Spin}(n+1)) \cong \pi_1(\text{Spin}(n+1)) = 0$. However, this just amounts to the following more direct argument. Choose lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ and represent each α_i by an embedding $\phi_i: S^2 \rightarrow W$. Then

$$w_k(\phi_i^*TW) = \phi_i^* w_k(TW) = 0$$

for $k = 1, 2$, since W is spin. The remainder of the proof proceeds as in Theorem 2.12. $\square$

In the main construction in Chapter ?? we will invoke Theorem 2.12 to obtain a PSC-metric, so let us conclude this section by discussing some basic results for identifying totally non-spin manifolds.

Lemma 2.13. (i) *The product of a totally non-spin manifold with any non-empty manifold is totally non-spin.*

- (ii) A manifold containing a totally non-spin submanifold of codimension 0 is totally non-spin.
- (iii) If M is totally non-spin and of dimension $n \geq 3$, then $M - D^n$ is totally non-spin for any embedded disk $D^n \subseteq \text{int}(M)$ in the interior of M .

Proof. For (i), let M_1 and M_2 be manifolds with M_1 totally non-spin. Then the universal cover of the product $M_1 \times M_2$ is given by the product $\tilde{M}_1 \times \tilde{M}_2$ of the respective universal covers. Denote by

$$\pi_1: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_1, \quad \pi_2: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_2$$

the projections and let $\iota: \tilde{M}_1 \rightarrow \tilde{M}_1 \times \tilde{M}_2$ be the inclusion for some choice of basepoint in $\tilde{M}_2$. We compute

$$\begin{aligned} \iota^* w_2(T(\tilde{M}_1 \times \tilde{M}_2)) &= \iota^* w_2(\pi_1^* T\tilde{M}_1 \oplus \pi_2^* T\tilde{M}_2) \\ &= w_2((\pi_1 \circ \iota)^* T\tilde{M}_1 \oplus (\pi_2 \circ \iota)^* T\tilde{M}_2) \\ &= w_2(T\tilde{M}_1) \neq 0, \end{aligned}$$

where we used that $\pi_1 \circ \iota$ is the identity and that $\pi_2 \circ \iota$ is constant, so that $(\pi_2 \circ \iota)^* T\tilde{M}_2$ is trivial. Thus $w_2(T(\tilde{M}_1 \times \tilde{M}_2)) \neq 0$.

For (ii), let M be a manifold and $N \subseteq M$ some totally non-spin codimension 0 submanifold. Let $\iota: N \rightarrow M$ be the inclusion and consider its lift to the universal covers:

$$\begin{array}{ccc} \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \\ \downarrow & & \downarrow \\ N & \xrightarrow{\iota} & M \end{array}$$

Then the differential $d\tilde{\iota}: T\tilde{N} \rightarrow T\tilde{M}$ is an isomorphism on fibers, so that $\tilde{\iota}^* T\tilde{M} \cong T\tilde{N}$. Therefore $\tilde{\iota}^* w_2(T\tilde{M}) = w_2(T\tilde{N}) \neq 0$ and so $w_2(T\tilde{M}) \neq 0$.

For (iii), let $M_0 := M - D^n$. By the Seifert-van-Kampen Theorem, the inclusion $i: M_0 \rightarrow M$ induces an isomorphism on π_1 . Therefore the universal cover $\tilde{M}_0$ of M_0 is given by the restriction of the universal cover $\tilde{M}$ of M to M_0 . Hence, $T\tilde{M}_0 = i^* T\tilde{M}$. By a Mayer-Vietoris sequence argument, i is injective on $H^2(-; \mathbb{Z}_2)$ and consequently we obtain

$$w_2(T\tilde{M}_0) = i^* w_2(T\tilde{M}) \neq 0.$$

□

3. Enlargeability

In this chapter we discuss the concept of enlargeability, which is the central idea that the results in this thesis are concerned with. The following definition is, for example, adopted in [LM89; BH10; CS21].

Definition 3.1. A closed oriented Riemannian n -manifold M is called enlargeable if for all $\varepsilon > 0$ there exists a Riemannian covering $\hat{M} \rightarrow M$ together with an ε -Lipschitz map $\hat{M} \rightarrow S^n$, which is constant outside a compact subset and of non-zero degree.

Note that this notion does not depend on the particular choice of Riemannian metric on M , since for closed manifolds all metrics are in bi-Lipschitz correspondence. Intuitively, the covering $\hat{M}$ can be pictured as an “expansion” of M itself (think, for example, of the usual sketches of coverings of S^1). Enlargeability now means that this expansion can be chosen arbitrarily large, so that it can be wrapped as tightly around a sphere as one wishes. Naively, one might expect that for an enlargeable manifold M the universal cover $\tilde{M}$ - which in some sense is the “largest” covering - is hyperspherical, meaning that it admits an ε -Lipschitz map $\tilde{M} \rightarrow S^n$, which is constant outside a compact subset and of non-zero degree for arbitrarily small $\varepsilon > 0$. However, Brunnbauer and Hanke have constructed counterexamples to this naive expectation [BH10, Thm. 1.4].

The simplest example of an enlargeable manifold is the n -torus T^n , since it is covered by $\mathbb{R}^n$, which is clearly hyperspherical. More generally, the same holds for any closed manifold admitting a metric of non-positive sectional curvature by the Cartan-Hadamard Theorem. Non-enlargeable manifolds are, for example, all simply connected manifolds, since all covers are trivial in this case.

The central fact about enlargeable manifolds is that they do not admit PSC-metrics.

Theorem 3.2. *An enlargeable manifold does not admit a metric of positive scalar curvature.*

This is the result that originally motivated the definition of enlargeability by Gromov and Lawson [GL80b], where they first showed that enlargeability is an obstruction to the existence of a PSC-metric, under the additional assumption that the coverings $\hat{M}$ in Definition 3.1 are finite and spin. Note that T^n is still enlargeable even under these additional assumptions. In fact, using their result about enlargeability, Gromov and Lawson gave the first general proof of the fact that T^n does not admit a PSC-metric for any dimension n . In a later paper, they were able to drop the finiteness assumption on $\hat{M}$ [GL83]. Both proofs are based on techniques involving the Dirac operator, which requires the existence of a spin structure, as briefly discussed in Chapter 2. So it was unclear whether and how these results carry over to the non-spin case. Rather recently, however, Cecchini and Schick resolved this issue [CS21]. Their proof is based on another fundamental method for obtaining obstructions to positive scalar curvature: *minimal hypersurfaces*. This approach is due to Schoen and Yau, and the core observation is the following. Suppose a closed manifold M^n admits a PSC-metric and $N^{n-1} \subseteq M$ is a local minimum of the volume functional. Then N also admits a PSC-metric [SY79, Proof of Thm. 1], [Lee19, Prop. 2.25]. So, to obstruct the existence of a PSC-metric on M , we can try to obstruct the existence of a PSC-metric on N . This idea can then be used inductively to reduce the problem to one of low dimension, where more methods are available (for example, the

3. Enlargeability

Gauß-Bonnet Theorem in dimension 2). The problem with this approach is the existence of such a minimal hypersurface N . In general, existence results for minimal hypersurfaces only hold up to dimension $n \leq 7$. However, at least in certain cases, Schoen and Yau proposed methods for overcoming this dimensional restriction [SY17], which is what Cecchini and Schick ultimately relied on to extend Theorem 3.2 to the non-spin case.

The following lemma now provides two easy ways to construct new enlargeable manifolds out of given ones.

Lemma 3.3. (i) *The product of enlargeable manifolds is enlargeable.*

(ii) *A closed oriented manifold admitting a non-zero degree map onto an enlargeable manifold is enlargeable.*

Proof. For ??, let M^m and N^n be enlargeable. For $\varepsilon > 0$, choose Riemannian coverings $p: \hat{M} \rightarrow M$ and $q: \hat{N} \rightarrow N$ as well as ε -contracting maps $f: \hat{M} \rightarrow S^m$ and $g: \hat{N} \rightarrow S^n$ of non-zero degree. Then

$$p \times q: \hat{M} \times \hat{N} \rightarrow M \times N$$

is a Riemannian covering, where both manifolds are equipped with the corresponding product metric. Moreover,

$$f \times g: \hat{M} \times \hat{N} \rightarrow S^m \times S^n$$

defines an ε -contracting map of degree $\deg(f \times g) = \deg(f)\deg(g) \neq 0$. Now consider the map

$$\phi: S^m \times S^n \rightarrow S^m \wedge S^n \cong S^{m+n},$$

given by projection onto the smash product

$$S^m \wedge S^n := S^m \times S^n / (S^m \times * \cup * \times S^n),$$

which is well known to be homeomorphic to the sphere S^{m+n} . The map ϕ is easily seen to be of degree ± 1 (depending on the chosen orientations) and, after a slight perturbation, we may also assume it to be smooth. Moreover, because $S^m \times S^n$ is closed, it follows that ϕ is a λ -Lipschitz map for some constant $\lambda > 0$. Therefore, $\phi \circ (f \times g)$ is a $\lambda\varepsilon$ -Lipschitz map of non-zero degree, and since ε was chosen arbitrarily, the statement follows.

For ??, let M^m be enlargeable and let $h: P \rightarrow M$ be some non-zero degree map. Again, given $\varepsilon > 0$, choose a Riemannian covering $p: \hat{M} \rightarrow M$ as well as an ε -contracting map $f: \hat{M} \rightarrow S^m$. Define a covering $\pi: \hat{P} \rightarrow P$ as the pullback covering of p under h , that is, π fits into a pullback diagram of the form

$$\begin{array}{ccc} \hat{P} & \xrightarrow{H} & \hat{M} \\ \downarrow \pi & & \downarrow p \\ P & \xrightarrow{h} & M \end{array}$$

where H is some suitable smooth map. Since $\hat{M}$ and P are closed, the same holds for $\hat{P}$. Moreover, h is μ -Lipschitz for some constant $\mu > 0$, because P is closed. Consequently, H is also μ -Lipschitz, given that both π and p are local isometries. For the degree of H we compute

$$\deg(p)\deg(H) = \deg(p \circ H) = \deg(h \circ \pi) = \deg(h)\deg(\pi) \neq 0,$$

using the fact that $\deg(h) \neq 0$ by assumption and that the degree of a covering is given by the cardinality of the fiber. We conclude that H is of non-zero degree, so that $f \circ H: \hat{P} \rightarrow S^m$ is a $\mu\varepsilon$ -Lipschitz map of non-zero degree. Again, since ε was chosen arbitrarily, the statement follows. $\square$

From Part ?? of the above lemma it follows immediately that enlargeability is a homotopy invariant, since a homotopy equivalence of connected closed oriented manifolds has degree ± 1 . In fact, much more is true, namely that enlargeability is a homological invariant in the following sense.

Theorem ([BH10]). *Let M be a closed oriented n -manifold and let $\phi: M \rightarrow B\pi_1(M)$ be the classifying map of the universal cover. Then M is enlargeable if and only if $\phi_*[M] \in H_n(B\pi_1(M); \mathbb{Q})$ does not lie in the rational vector subspace*

$$H_n^{\text{small}}(B\pi_1(M); \mathbb{Q}) \subseteq H_n(B\pi_1(M); \mathbb{Q})$$

of “small” classes, i.e., $\phi_[M]$ is “large”.*

In [BH10], the authors showed this by giving a general definition of “small” and “large” homology classes of a simplicial complex X with finitely generated fundamental group, such that the assignment $X \mapsto H_n^{\text{small}}(X; \mathbb{Q})$ is functorial for continuous maps $f: X \rightarrow Y$ inducing an epimorphism on π_1 , that is, $f_*: H_n(X; \mathbb{Q}) \rightarrow H_n(Y; \mathbb{Q})$ maps small classes to small classes. Moreover, if f happens to induce an isomorphism on π_1 , then f_* also maps large classes to large classes. This general definition then specializes to the fact that a closed oriented manifold M is enlargeable if and only if the fundamental class $[M]$ is large, that is, $[M] \notin H_n^{\text{small}}(M; \mathbb{Q})$. Combining these observations, the theorem follows. Furthermore, this argument shows that the notion of enlargeability can be understood purely in terms of simplicial complexes. For details, we refer the reader to [BH10].

4. Main Construction

Recall the following question discussed in the introduction.

Question A. Do there exist circle bundles $E \rightarrow M$ over an enlargeable manifold M , such that the total space E is non-enlargeable? Moreover, do there exist 2-dimensional real vector bundles $V \rightarrow M$, where V admits a complete metric of uniformly positive scalar curvature?

We begin this chapter with an alternative approach to the one presented in [KS25] for constructing examples addressing the first part of the question. We subsequently build on this construction to obtain an affirmative answer to the second part of the question.

4.1. Construction of Circle Bundles with PSC

In this section we construct examples of circle bundles $E \rightarrow M$ over enlargeable manifolds M in all dimensions $\dim(M) \geq 4$, such that E admits a PSC-metric. In particular E is non-enlargeable by Theorem 3.2. Let us first describe the strategy behind the construction. The existence of a PSC-metric on E will be established as an application of Theorem 2.12, restated here for convenience:

Theorem 2.12. *Let W^{n+1} be an oriented cobordism from M_0 to M_1 , $n \geq 5$. Suppose that M_1 is a connected totally non-spin manifold, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary.*

In order to apply this in our case, we make the following two observations, which hold true for any circle bundle $E \rightarrow M$ over a closed orientable manifold M .

- (i) There exists an oriented nullbordism of E , namely the total space of the associated disk bundle $DE \rightarrow M$,
- (ii) The inclusion $E \cong \partial DE \rightarrow DE$ induces an isomorphism on π_1 if and only if the fibers of E are contractible, that is the inclusion $S^1 \rightarrow E$ is trivial on π_1 .

We have already observed in Example (iii) that DE is a nullbordism of E . Moreover, the structure group of $DE \rightarrow M$ acts on fibers by orientation preserving diffeomorphisms. Thus any orientation on M gives rise to an orientation on DE , which is uniquely determined by requiring local trivialisations to be orientation preserving. For (ii) note that $DE \rightarrow M$ is a homotopy equivalence. Hence the diagram

$$\begin{array}{ccc} E & \longrightarrow & DE \\ & \searrow & \swarrow \\ & M & \end{array} \quad \simeq$$

shows that $E \rightarrow DE$ induces an isomorphism on π_1 if and only if $E \rightarrow M$ does. Then the claim follows from the long exact sequence of homotopy groups:

$$\pi_1(S^1) \longrightarrow \pi_1(E) \longrightarrow \pi_1(M) \longrightarrow \pi_0(S^1)$$

4. Main Construction

With these observations in mind, we have to construct circle bundles $E \rightarrow M$ with the following properties.

- (I) M is enlargeable, orientable and connected,
- (II) E is totally non-spin,
- (III) The inclusion $S^1 \rightarrow E$ is trivial on π_1 .

We define M as the connected sum of three closed manifolds. Let N^{2n} be any $2n$ -dimensional orientable totally non-spin manifold. For example take $N := \mathbb{C}P^2 \times T^{2n-4}$, which is totally non-spin by Lemma 2.13 and Example (ii). Then consider

$$M := \mathbb{C}P^n \# T^{2n} \# N.$$

Intuitively the bundle $E \rightarrow M$ will now look as follows. It is trivial over the summands T^{2n} and N , and is given by the canonical bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$ over the $\mathbb{C}P^n$ summand (or rather the restriction of this bundle to $\mathbb{C}P^n - D^{2n} \subseteq M$). Schematically:

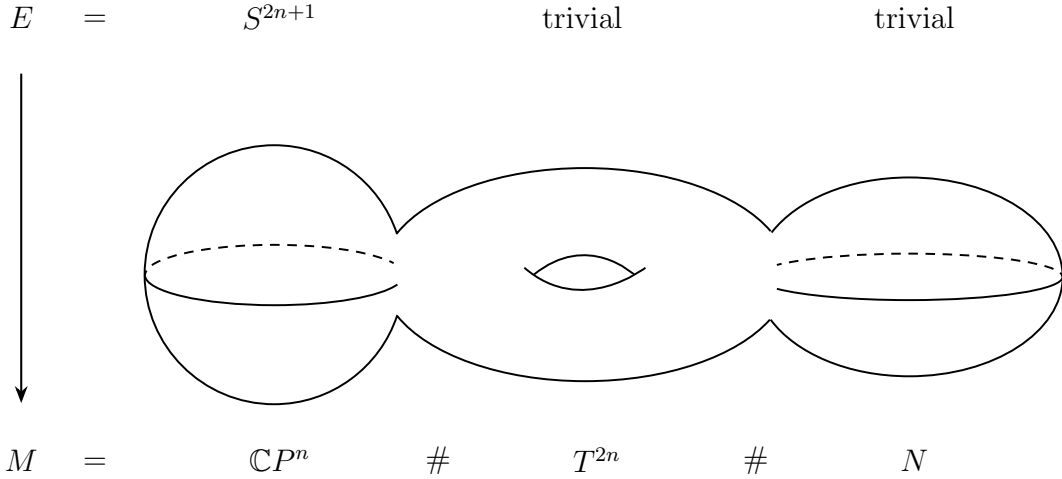


Figure 4.1.: Sketch of the constructed circle bundle

The presence of the Torus summand then corresponds to property (I), the trivial bundle over N to property (II) and the S^{2n+1} over $\mathbb{C}P^n$ to property (III). We will now construct this bundle rigourously and subsequently discuss the proof of properties (I) - (III) in more detail. Let $\alpha \in H^2(\mathbb{C}P^n)$ be the generator given boundary the first chern class of the tautological complex line bundle $\tau_1^n \rightarrow \mathbb{C}P^n$. Implicitly all homology and cohomology groups in this chapter are taken with $\mathbb{Z}$ -coefficients. Consider the isomorphism

$$\phi: H^2(\mathbb{C}P^n) \oplus H^2(T^{2n}) \oplus H^2(N) \rightarrow H^2(M)$$

and let $c_1 \in H^2(M)$ be the class corresponding to $(\alpha, 0, 0)$ under this isomorphism. By Theorem 1.19 there exists a unique complex line bundle $L \rightarrow M$ with first chern class $c_1(L) = c_1$. Now define $E \rightarrow M$ as the S^1 -principal bundle associated to this line bundle. We claim that E is of the form as described above. To this end we will show that $L \rightarrow M$ is trivial over the summands T^{2n} and N , and is given by the restriction of $\tau_1^n \rightarrow \mathbb{C}P^n$ to $\mathbb{C}P^n - D^{2n} \subseteq M$ over the $\mathbb{C}P^n$ summand. By Example 1.7 (ii) the associated principal bundle of $\tau_1^n \rightarrow \mathbb{C}P^n$ is precisely $S^{2n+1} \rightarrow \mathbb{C}P^n$, such that the claim then follows from the naturality of changing the fiber, see Remark 1.6. To establish the statements made about $L \rightarrow M$ let

$$i: N - D^{2n} \rightarrow M, \quad i_1: N - D^{2n} \rightarrow N$$

be the inclusions. Then

$$\begin{array}{ccc} H^2(M) & \xrightarrow{i^*} & H^2(N - D^{2n}) \\ \cong \uparrow \phi & & \cong \uparrow i_1^* \\ H^2(\mathbb{C}P^n) \oplus H^2(T^{2n}) \oplus H^2(N) & \xrightarrow{\text{proj.}} & H^2(N) \end{array}$$

commutes and therefore $c_1(i^*L) = i^*(c_1) = 0$. Again using Theorem 1.19 we conclude that i^*L is trivial. An analogous argument shows triviality of L over the T^{2n} summand. Similarly, if we denote by

$$j: \mathbb{C}P^n - D^{2n} \rightarrow M, \quad j_1: \mathbb{C}P^n - D^{2n} \rightarrow \mathbb{C}P^n$$

the inclusions, the diagram

$$\begin{array}{ccc} H^2(M) & \xrightarrow{j^*} & H^2(\mathbb{C}P^n - D^{2n}) \\ \cong \uparrow \phi & & \cong \uparrow j_1^* \\ H^2(\mathbb{C}P^n) \oplus H^2(T^{2n}) \oplus H^2(N) & \xrightarrow{\text{proj.}} & H^2(\mathbb{C}P^n) \end{array}$$

commutes and thus $c_1(j^*L) = j^*(c_1) = j_1^*(\alpha) = c_1(j_1^*\tau_1^n)$. Hence, applying Theorem 1.19 once more shows that j^*L and $j_1^*\tau_1^n$ are isomorphic bundles. This concludes the construction of E . Let us now verify properties (I) - (III).

Proof of property (I). M is connected and orientable, as each summand is. Moreover, since the collapse map $M \rightarrow T^{2n}$ has degree 1 and T^{2n} is enlargeable, it follows from Lemma 3.3 that M is also enlargeable. $\square$

Proof of property (II). Let $i: N - D^{2n} \rightarrow M$ be the inclusion. We have established above, that

$$i^*E \cong (N - D^{2n}) \times S^1.$$

From Lemma 2.13 (i) and (iii) we conclude that $(N - D^{2n}) \times S^1$ is totally non-spin. Therefore E contains a totally non-spin submanifold of codimension 0 and so Lemma 2.13 (ii) implies that E is also totally non-spin. $\square$

Proof of property (III). Intuitively we can contract the fibers over the S^{2n+1} contained in E . One way to formalize this is the following. Consider the following commutative diagram, where the rows are taken from the long exact sequences of homotopy groups of the respective fibrations and the vertical maps are induced by inclusions:

$$\begin{array}{ccccc} \pi_2(M) & \xrightarrow{\partial_1} & \pi_1(S^1) & \longrightarrow & \pi_1(E) \\ \uparrow & & \uparrow \text{id} & & \uparrow \\ \pi_2(\mathbb{C}P^n - D^{2n}) & \xrightarrow{\partial_2} & \pi_1(S^1) & \longrightarrow & \pi_1(j^*E) \\ \cong \downarrow & & \downarrow \text{id} & & \downarrow \\ \pi_2(\mathbb{C}P^n) & \xrightarrow{\partial_3} & \pi_1(S^1) & \longrightarrow & \pi_1(S^{2n+1}) \end{array}$$

The vertical map on the lower left is an isomorphism, because both $\mathbb{C}P^n$ and $\mathbb{C}P^n - D^{2n}$ are simply connected and so by the Hurewicz Theorem

$$\begin{array}{ccc} \pi_2(\mathbb{C}P^n - D^{2n}) & \longrightarrow & \pi_2(\mathbb{C}P^n) \\ \downarrow \cong & & \downarrow \cong \\ H_2(\mathbb{C}P^n - D^{2n}) & \longrightarrow & H_2(\mathbb{C}P^n) \end{array}$$

4. Main Construction

commutes and the vertical maps are isomorphisms. Since the bottom horizontal map is an isomorphism the same holds for the top horizontal map. We now claim that ∂_1 is surjective. For this it suffices to show surjectivity of ∂_2 , which in turn is equivalent to surjectivity of ∂_3 . But this is clear by exactness of the rows and $\pi_1(S^{2n+1}) = 0$. Consequently

$$\pi_1(S^1) \rightarrow \pi_1(E)$$

is the zero map. $\square$

The above construction now gives examples for circle bundles with positive scalar curvature over enlargeable manifolds M in all even dimensions $\dim(M) \geq 4$. We argue as in [KS25] to also obtain examples in odd dimensions as follows. Let $E \rightarrow M$ be as above and let $\pi: M \times S^1 \rightarrow M$ be the projection. Then by Example 2.2 the pullback bundle $\pi^*E \cong E \times S^1$ admits a metric of positive scalar curvature. Furthermore the base space $M \times S^1$ is still enlargeable by Lemma 3.3. Naturally, the question arises as to what happens in dimensions $\dim(M) \leq 3$. In this case it turns out that no circle bundle $E \rightarrow M$ over an enlargeable manifold M admits a metric of positive scalar curvature. This was shown in [KS25, Theorem B, Remark 2].

4.2. Necessity of the Enlargeability Assumption in Gromov's Decay Theorem

In this section we will use the preceeding results to construct examples of 2-dimensional real vector bundles $V \rightarrow M$ over enlargeable manifolds M , such that V admits a complete metric of uniformly positive scalar curvature.

Let $E \rightarrow M$ be a circle bundle as constructed in the previous section. The existence of a PSC-metric on E followed from an application of Theorem 2.12. However, we did not make use of the full strength of this theorem. Not only do we obtain a PSC-metric on E but also a PSC-metric on the nullbordism DE , which is of product form near the boundary. This observation allows for the following construction. Let g be such a PSC-metric on DE and denote the induced metric on E by g_E . Choose some collar neighbourhood $\iota: E \times [0, 1) \rightarrow DE$ with respect to which g is in product form, i.e.

$$\iota^*g = g_E + dt^2.$$

Moreover, consider the Riemannian manifold $(E \times [1, \infty), g_E + dt^2)$ equipped with the identity as a collar neighbourhood. We can glue this manifold with (DE, g) along the common boundary E to define another Riemannian manifold

$$(V', h) := (E \times [1, \infty) \cup_E DE, (g_E + dt^2) \cup_E g).$$

We explicitly mention that the smooth structure on V' is defined with respect to the chosen collar neighbourhoods ι and $\text{id}_{E \times [1, \infty)}$, such that h is indeed smooth. Note that the metrics on both $E \times [1, \infty)$ and DE are complete, therefore h is also complete. Furthermore, g_E is a PSC-metric, hence the scalar curvature of $g_E + dt^2$ is uniformly positive by compactness of E and Example 2.2 (iv). The same holds for the scalar curvature of g . Consequently, scal_h is uniformly positive as well. Now consider the map $p_0: E \times [1, \infty) \rightarrow M$ defined as the composition of the projection $E \times [1, \infty) \rightarrow E$ and the bundle projection $p: E \rightarrow M$. Together with the bundle projection $q: DE \rightarrow M$ we set

$$\pi := p_0 \cup_E q: V' \rightarrow M,$$

which we will show to be a rank 2 vector bundle. However, in general we can only expect π to be a continuous map as the collar neighbourhood ι contains no information about

4.2. Necessity of the Enlargeability Assumption in Gromov's Decay Theorem

the bundle structure on DE . To resolve this subtlety, we proceed as follows, being quite precise about the details. Choose trivialisations

$$\phi_\alpha: U_\alpha \times S^1 \rightarrow p^{-1}(U_\alpha), \quad \phi_\beta: U_\beta \times S^1 \rightarrow p^{-1}(U_\beta)$$

of $E \rightarrow M$ with smooth transition function $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow S^1$. By definition there are then trivialisations

$$\psi_\alpha: U_\alpha \times D^2 \rightarrow q^{-1}(U_\alpha), \quad \psi_\beta: U_\beta \times D^2 \rightarrow q^{-1}(U_\beta)$$

of $DE \rightarrow M$ with the same transition function $\tau_{\alpha\beta}$. Let $B \subseteq \mathbb{R}^2$ be the open ball of radius 1 and consider the diffeomorphism

$$\mu: \mathbb{R}^2 - B \rightarrow S^1 \times [1, \infty), \quad w \mapsto \left(\frac{w}{|w|}, |w| \right),$$

which is equivariant under the canonical S^1 -actions. Then we can define a trivialisation

$$\theta_\alpha: U_\alpha \times (\mathbb{R}^2 - B) \xrightarrow{\text{id} \times \mu} U_\alpha \times S^1 \times [1, \infty) \xrightarrow{\phi_\alpha \times \text{id}} p^{-1}(U_\alpha) \times [1, \infty) = p_0^{-1}(U_\alpha)$$

of $E \times [1, \infty)$ and similarly $\theta_\beta: U_\beta \times (\mathbb{R}^2 - B) \rightarrow p_0^{-1}(U_\beta)$. By construction, the transition function of these trivialisations is again $\tau_{\alpha\beta}$. The maps θ_α and ψ_α now glue together to a *homeomorphism*

$$\xi_\alpha: U_\alpha \times \mathbb{R}^2 = (U_\alpha \times (\mathbb{R}^2 - B)) \cup (U_\alpha \times D^2) \xrightarrow{\theta_\alpha \cup \psi_\alpha} p_0^{-1}(U_\beta) \cup q^{-1}(U_\beta) = \pi^{-1}(U_\alpha).$$

Likewise we obtain a homeomorphism $\xi_\beta: U_\beta \times \mathbb{R}^2 \rightarrow \pi^{-1}(U_\beta)$. From the discussion about the transition functions we conclude

$$(\xi_\beta^{-1} \circ \xi_\alpha)(x, w) = (x, \tau_{\alpha\beta}(x) \cdot w).$$

Observe that this map is smooth. Therefore, the collection of maps constructed in the same manner as ξ_α define a smooth structure on V' . This structure may not agree with the one previously defined with respect to the collar neighbourhoods, so for clarity let us refer to the topological manifold V' equipped with this new smooth structure as V . Then the map $\pi: V \rightarrow M$ is clearly smooth and carries the structure of a *smooth* rank 2 vector bundle. To now obtain a complete metric on V with uniformly positive scalar curvature we recall the following. The smooth structure on V' (defined via the collar neighbourhoods) is up to diffeomorphism the unique smooth structure on the topological manifold V' , such that the topological embeddings

$$E \times [1, \infty) \rightarrow V', \quad DE \rightarrow V'$$

are smooth embeddings [hcbord, Theorem 1.4]. Note that the inclusion $E \times [1, \infty) \rightarrow V$ is locally given by the inclusion

$$U_\alpha \times (\mathbb{R}^2 - B) \rightarrow U_\alpha \times \mathbb{R}^2$$

with respect to the smooth trivialisations θ_α and ξ_α . Similarly the inclusion $DE \rightarrow V$ is locally given by the inclusion

$$U_\alpha \times D^2 \rightarrow U_\alpha \times \mathbb{R}^2$$

with respect to ψ_α and ξ_α . Hence we conclude from the uniqueness statement above that there exists a diffeomorphism $f: V \rightarrow V'$. Then the metric f^*h on V is complete and of uniformly positive scalar curvature.

5. Further Questions

5.1. Problems with Extending the Construction to the Spin Case

The base spaces M of the circle bundles $E \rightarrow M$ constructed in the previous Chapter are all totally non-spin manifolds. As we have seen in Chapter ?? many constructions in positive scalar curvature geometry require the existence of a spin structure, so it may also be desirable to get examples of such circle bundles over spin manifolds. However, our methods do not straightforwardly extend to the spin case. Let us discuss the difficulties that arise.

Suppose we are given an S^1 -principal bundle $\pi: E \rightarrow M$ over a spin manifold M (in particular M is orientable). First of all we cannot conclude the existence of a PSC-metric on E from Theorem 2.12 anymore, since the total space E is itself a spin manifold. The reason is the following. There exists a splitting

$$TE \cong HE \oplus VE,$$

of the tangent bundle TE of E into a *vertical* bundle VE and a *horizontal* bundle HE . The bundle VE is simply the kernel of the differential $d\pi: TE \rightarrow TM$ and we can then choose HE as any complementary subbundle (for example by choosing a fiber metric on TE). By construction $d\pi: HE \rightarrow TM$ is an isomorphism on fibers, such that $HE \cong \pi^*TM$. In our case VE is one-dimensional, such that $w_2(VE) = 0$ and therefore

$$w_2(TE) = w_1(HE) \smile w_1(VE) + w_2(HE) = \pi^*w_1(TM) \smile w_1(VE) + \pi^*w_2(TM) = 0.$$

So instead we are inclined to use the spin analogue of Theorem 2.12 to conclude the existence of a PSC-metric on E , namely:

Theorem 2.11. *Let W^{n+1} be a cobordism from M_0 to M_1 , $n \geq 5$, such that W is spin. Suppose that M_1 is connected, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary. Proceeding as before we would like to take the disk bundle DE as a nullbordism of E and show:*

(I') DE is spin,

(II') The inclusion $S^1 \rightarrow E$ is trivial on π_1 .

However, these properties contradict themselves. To see this we make the following observations. Let $L \rightarrow M$ be the complex line bundle associated to $E \rightarrow M$ and consider the inclusion $\iota: DE \rightarrow L$. Since the differential $d\iota: TDE \rightarrow TL$ is an isomorphism on fibers we conclude $TDE \cong \iota^*TL$. Thus

$$\iota^*w_2(TL) = w_2(TDE) = 0$$

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by property (I'). But ι is a homotopy equivalence so consequently $w_2(TL) = 0$. Let $p: L \rightarrow M$ be the bundle projection. As before we have a splitting

$$TL \cong HL \oplus VL$$

into a vertical and horizontal subbundle with $HL \cong p^*TM$. For vector bundles it is a standard fact that $VL = \ker(dp) \cong p^*E$, hence

$$0 = w_2(TL) = \pi^*w_2(HL) + \pi^*w_1(HL) \smile p^*w_1(L) + p^*w_2(L) = p^*w_2(L).$$

Again we conclude $w_2(E) = 0$ since p is a homotopy equivalence. Therefore the first chern class $c_1(L)$ lies in the kernel of the coefficient homomorphism

$$H^2(M; \mathbb{Z}) \rightarrow H^2(M; \mathbb{Z}_2)$$

by Proposition 1.17. In other words, $c_1(L)$ can be written as $c_1(L) = 2\alpha$ for some $\alpha \in H^2(M; \mathbb{Z})$. To derive a contradiction from this we claim that the following diagram commutes up to sign

$$\begin{array}{ccc} \pi_2(M) & \xrightarrow{\partial} & \pi_1(S^1) \\ \downarrow & & \downarrow \cong \\ H_2(M; \mathbb{Z}) & \xrightarrow{\langle c_1(L), - \rangle [S^1]} & H_1(S^1; \mathbb{Z}) \end{array}$$

where ∂ is the boundary map from the long exact sequence of homotopy groups for the fibration $E \rightarrow M$ and the vertical maps are given by the Hurewicz homomorphism. Moreover $[S^1]$ denotes the fundamental class of S^1 . Granting this claim for now, it then follows from $c_1(L) = 2\alpha$ that $\partial: \pi_2(M) \rightarrow \pi_1(S^1)$ cannot be surjective and consequently $\pi_1(S^1) \rightarrow \pi_1(E)$ is non-zero by exactness. Hence a bundle satisfying both (I') and (II') does not exist. To show commutativity of the above square let $\tau_1 \rightarrow \mathbb{C}P^\infty$ be the universal complex line bundle (compare Example 1.14 (vi)) and let $f: M \rightarrow \mathbb{C}P^\infty$ be a classifying map for $L \rightarrow M$. Then essentially commutativity is due to the fact that the corresponding square for the *universal* principal bundle commutes and all maps are compatible with the classifying map f . To make this precise we extend the diagram as follows

$$\begin{array}{ccccc} & & \partial & & \\ & & \cong & & \\ \pi_2(\mathbb{C}P^\infty) & \xrightarrow{\quad} & \pi_1(S^1) & & \\ & \nwarrow f_* & \downarrow \text{id} & \nearrow & \\ & \pi_2(M) & \xrightarrow{\partial} & \pi_1(S^1) & \\ & \downarrow & \downarrow \cong & \downarrow & \\ & H_2(M; \mathbb{Z}) & \xrightarrow{\langle c_1(L), - \rangle [S^1]} & H_1(S^1; \mathbb{Z}) & \\ & \nwarrow f_* & \downarrow \text{id} & \nearrow & \\ H_2(\mathbb{C}P^\infty; \mathbb{Z}) & \xrightarrow{\quad} & H_1(S^1; \mathbb{Z}) & & \\ & \nwarrow \langle c_1(\tau_1), - \rangle [S^1] & \nearrow & & \end{array}$$

(2) (3) (1) (4)

Again the top horizontal map is the boundary map from the long exact sequence for the fibration $ES^1 \rightarrow \mathbb{C}P^\infty$ and the outer vertical maps are given by the Hurewicz homomorphism. Since $\mathbb{C}P^\infty$ is simply connected $\pi_2(\mathbb{C}P^\infty) \rightarrow H_2(\mathbb{C}P^\infty; \mathbb{Z})$ is an isomorphism. Moreover, the boundary map $\partial: \pi_2(\mathbb{C}P^\infty) \rightarrow \pi_1(S^1)$ is an isomorphism by Example 1.14 (i). Clearly (1) commutes. Furthermore (2) commutes by naturality of the long exact

5.2. A Question about the Homotopy Groups of the Space of PSC-Metrics

sequence of homotopy groups and (3) commutes by naturality of the Hurewicz homomorphism. For commutativity of (4) we compute for $a \in H_2(M; \mathbb{Z})$:

$$\langle c_1(\tau_1), f_*a \rangle = \langle f^*c_1(\tau_1), a \rangle = \langle c_1(L), a \rangle,$$

using the fact that f is a classifying map in the last equation. Finally we have to show commutativity of the outer square. From the universal coefficient theorem we obtain that

$$H^2(\mathbb{C}P^\infty; \mathbb{Z}) \rightarrow \text{Hom}(H_2(\mathbb{C}P^\infty; \mathbb{Z}), \mathbb{Z}), \quad \beta \mapsto \langle \beta, - \rangle$$

is an isomorphism. By definition $c_1(\tau_1) \in H^2(\mathbb{C}P^\infty; \mathbb{Z})$ is a generator, thus $\langle c_1(\tau_1), - \rangle$ is a generator of $\text{Hom}(H_2(\mathbb{C}P^\infty; \mathbb{Z}), \mathbb{Z})$ and consequently evaluates to ± 1 on a generator of $H_2(\mathbb{C}P^\infty; \mathbb{Z})$. From this we see that

$$\langle c_1(\tau_1), - \rangle [S^1]: H_2(\mathbb{C}P^\infty; \mathbb{Z}) \rightarrow H_2(S^1; \mathbb{Z})$$

is an isomorphism. So all maps in the outer square are isomorphisms and, moreover, all groups are isomorphic to $\mathbb{Z}$. Hence the outer square commutes up to sign and the claim follows.

Thus, we have established that our previous methods for constructing a PSC-metric on $E \rightarrow M$ do not apply when M is a spin manifold. Nevertheless, examples answering Question ?? in the spin case do exist. These were obtained by Kumar and Sen [KS25] using more sophisticated methods, as we have briefly sketched in the introduction.

5.2. A Question about the Homotopy Groups of the Space of PSC-Metrics

In Chapter ?? we have discussed sufficient conditions for the the existence of a PSC-metric on a manifold. Given a manifold M , which admits a PSC-metric we can further ask how many of such metrics exist. More precisely the question should be whether any two PSC-metrics on M can be deformed into each other via a continuous path of PSC-metrics. Phrased differently, if we denote by $\mathcal{R}^+(M)$ the space of all PSC-metrics on M , equipped with the C^∞ -topology, this statement asks whether $\pi_0(\mathcal{R}^+(M))$ is trivial. For instance, this is always true for $M = S^2$. In fact $\mathcal{R}^+(S^2)$ is contractible and so in particular connected [??]. However, in higher dimensions this breaks down even for such simple spaces as spheres as demonstrated by the following theorem due to Carr.

Theorem 5.1. *For $n \geq 2$ the space $\mathcal{R}^+(S^{4n-1})$ has infinitely many path components.*

This already illustrates the complexity the space of PSC-metrics may exhibit. Going one step further, we can also investigate the higher homotopy groups of this space, essentially asking which *families* of PSC-metrics can be deformed into each other. One result in this direction is the following theorem due to Hanke, Schick and Steimle.

Theorem 5.2. *Let $k \in \mathbb{N}_0$. Then there exists a number $N(k)$, such that for every $n \geq N(k)$ and every spin manifold M of dimension $4n - k - 1$ admitting a PSC-metric g_0 , the homotopy group*

$$\pi_k(\mathcal{R}^+(M), g_0)$$

contains elements of infinite order if $k \geq 1$ and infinitely many distinct elements if $k = 0$.

Note that for $k = 0$ and $M = S^{4n-1}$, the statement above generalizes Theorem 5.1 quite substantially. Tying this discussion back to our results of Chapter ??, let us analyze the PSC-metric constructed on E more closely. A natural question is whether this metric is

5. Further Questions

invariant under the S^1 -action on E . We can answer this question in the negative due to a theorem of Bérard-Bergery, which states that a closed manifold with a free S^1 -action admits an S^1 -invariant PSC-metric if and only if its orbit space admits a PSC-metric; see [BerardBergery1983] and [Wiemeler2016]. But in our case $E/S^1 \cong M$ and M is enlargeable. Therefore it follows from Theorem 3.2 that M does not admit a PSC-metric. In view of the above discussion about the homotopy groups of the space of PSC-metrics we are thus lead to the following question.

Question B. Let $g \in \mathcal{R}^+(E)$ be the PSC-metric constructed in Chapter ?? . Does the map

$$S^1 \rightarrow \mathcal{R}^+(E), \quad \zeta \mapsto \zeta^* g$$

represent a non-trivial element in $\pi_1(\mathcal{R}^+(E), g)$?

Given the preceding observations, the most that can be said is that this loop is non-constant. However, it may of course still be contractible. At the time of writing, we do not know the answer to Question B and leave it as an open problem for future investigation.

A. Appendix

A.1. A Short Introduction to Surgery and Cobordisms

Surgery Theory provides a way to change the topology of a manifold in a controlled way. The basic idea is the simple observation, that a product $S^k \times S^{n-k-1}$ of spheres can be considered as the boundary of both $S^k \times D^{n-k}$ and of $D^{k+1} \times S^{n-k-1}$. This allows for the following construction. Let M^n be a smooth n -dimensional manifold and let $F: S^k \times D^{n-k} \rightarrow M$ be an embedding into the interior of M . Then we can remove the interior of the image of F resulting in a manifold

$$\tilde{M} := M - \text{int}(\text{im}(F)).$$

To $\tilde{M}$ we can attach $D^{k+1} \times S^{n-k-1}$ along the boundary component $F(S^k \times S^{n-k-1}) \subseteq \tilde{M}$ to obtain a manifold M' . More precisely we define M' as the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \xrightarrow{F|_{S^k \times S^{n-k-1}}} & \tilde{M} \\ \downarrow \text{incl.} & & \downarrow \\ D^{k+1} \times S^{n-k-1} & \longrightarrow & M' \end{array}$$

Up to diffeomorphism there exists a unique smooth structure on M' , such that the topological embeddings $\tilde{M} \rightarrow M'$ and $D^{k+1} \times S^{n-k-1} \rightarrow M'$ are smooth embeddings, see [Milnor1965, Theorem 1.4]. We say that the manifold M' is obtained from M by surgery along F . Intuitively we remove a thickened k -sphere from M and glue in its place a thickened $(n - k - 1)$ -sphere. Accordingly we refer to the surgery along F as a surgery of dimension k or simply a k -surgery.

A different, yet closely related concept, is the theory of cobordisms. Two closed n -dimensional manifolds M and M' are called *cobordant* if there exists a compact manifold W^{n+1} whose boundary is the disjoint union of M and M' . The manifold W is called a *cobordism* between M and M' . If $M' = \emptyset$ we also refer to W as a *nullbordism* and call M *nullbordant*. Since any closed manifold is cobordant to itself via the trivial cobordism $M \times [0, 1]$ and cobordisms can be glued together along common boundary components, the relation of cobordism defines an equivalence relation on the class of closed n -dimensional manifolds. One way to construct a new cobordism out of a given one is by means of *handle attachments*. Suppose we are given an embedding $\phi: S^k \times D^{n-k} \rightarrow \partial W$. Then we can form a new manifold W' by attaching a so called $(k + 1)$ -handle $D^{k+1} \times D^{n-k}$ to W along ϕ , that is we form the pushout

$$\begin{array}{ccc} S^k \times D^{n-k} & \xrightarrow{\phi} & W \\ \downarrow \text{incl.} & & \downarrow \\ D^{k+1} \times D^{n-k} & \longrightarrow & W' \end{array}$$

We also refer to $(k + 1)$ as the *index* of the handle. Again some care needs to be taken to define a smooth structure on W' , see [Milnor, Morse Theory]. It turns out that any cobordism W from M to M' can be obtained from the trivial cobordism $M \times [0, 1]$ by attaching finitely many handles [REFERENCE]. Now the connection between surgery

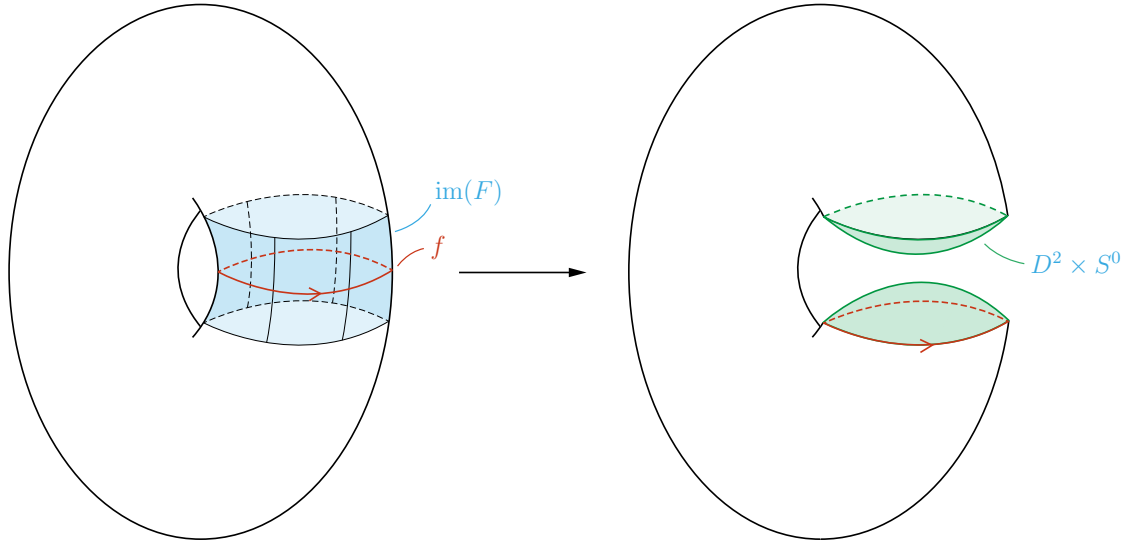


Figure A.1.: Performing a 1-surgery on the Torus to kill the homotopy class represented by f . The resulting space is a 2-sphere, where the corresponding class is now trivial.

theory and cobordisms is the following fundamental result, which can be shown by means of Morse theory.

Theorem A.1 (Milnor1965, Theorem 3.13 and Theorem 3.14 + other source!!). *Let M and M' be closed manifolds of the same dimension. Then M' is obtained from M by a surgery of dimension k if and only if there exists a cobordism W from M to M' containing only a single handle of index $k + 1$, that is W is obtained from the trivial cobordism $M \times [0, 1]$ by attaching a single $(k + 1)$ -handle.*

One common usecase of surgery is to modify a given manifold M in order to simplify its homotopy groups. Essentially given some embedding $f: S^k \rightarrow M$ we would like to modify the topology of M , such that f becomes contractible. We could just glue in a disk D^{k+1} along f , however this would destroy the manifold structure. So instead we “thicken” the embedded sphere by extending f to an embedding $F: S^k \times D^{n-k}$, remove this thickened sphere and glue in a whole family of disks $D^{k+1} \times S^{n-k-1}$, i.e. we perform a k -surgery. This process is usually referred to as “killing” the homotopy class $[f]$ by surgery. The fundamental obstruction to performing such a surgery is the existence of the embedding F , which exists if and only if the normal bundle of f is trivial. The following Proposition now formalizes this idea of killing homotopy classes by surgery.

Proposition A.2. *Let W^n be a smooth manifold with basepoint $w_0 \in W$, $M \subseteq \partial W$ a subset of the boundary with basepoint $x_0 \in M$, $2(k + 1) \leq n$ and $k \geq 1$.*

- (i) *Let $\alpha_1, \dots, \alpha_l \in \pi_k(W, w_0)$ be classes, which are represented by embeddings with trivial normal bundle. Then there exists a manifold W' obtained from W by l -many k -surgeries away from the point w_0 together with a homomorphism $\phi: \pi_1(W, w_0) \rightarrow \pi_1(W', w_0)$ and a ϕ -equivariant epimorphism*

$$\pi_k(W, w_0) \twoheadrightarrow \pi_k(W', w_0)$$

containing $\alpha_1, \dots, \alpha_l$ in its kernel. Moreover,

$$\pi_i(W, w_0) \cong \pi_i(W', w_0)$$

for all $i \leq k - 1$.

(ii) Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_k(W, M, x_0)$ be classes, which lift to classes $\alpha_1, \dots, \alpha_l \in \pi_k(W, x_0)$ under the map $\pi_k(W, x_0) \rightarrow \pi_k(W, M, x_0)$ induced by the inclusion. Again assume that $\alpha_1, \dots, \alpha_l$ are represented by embeddings with trivial normal bundle. Then there exists a manifold W' obtained from W by l -many k -surgeries together with a $\pi_1(M, x_0)$ -equivariant epimorphism

$$\pi_k(W, M, x_0) \twoheadrightarrow \pi_k(W', M, x_0)$$

containing $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ in its kernel. Moreover,

$$\pi_i(W, M, x_0) \cong \pi_i(W', M, x_0)$$

for all $1 \leq i \leq k-1$.

Remark A.3. If $\pi_k(W, w_0)$ is abelian (for example if $k \geq 2$) equivariance of the epimorphism $\pi_k(W, w_0) \twoheadrightarrow \pi_k(W', w_0)$ implies that it contains the entire $\mathbb{Z}[\pi_1(W, w_0)]$ -module generated by $\alpha_1, \dots, \alpha_l$ in its kernel. An analogous statement holds in the relative case.

In the proof of the Proposition we will make use of the following basic lemma.

Lemma A.4. *Let X be a topological space and let $H: S^k \times [0, 1] \rightarrow X$ be a homotopy from some map $f: S^k \rightarrow X$ to some map $g: S^k \rightarrow X$. Fix a basepoint $s_0 \in S^k$ and consider the path $\gamma(t) := H(s_0, t)$ from $x_0 := f(s_0)$ to $x_1 := g(s_0)$. Then the change of basepoint isomorphism*

$$b_\gamma: \pi_k(X, x_0) \rightarrow \pi_k(X, x_1)$$

induced by γ maps the class $[f]$ to the class $[g]$.

Proof. Represent the classes $[f]$ and $[g]$ by maps $f': D^k \rightarrow X$ and $g': D^k \rightarrow X$, which are constant on the boundary. Let $H': D^k \times [0, 1] \rightarrow X$ be the corresponding homotopy from f' to g' , such that $H'(S^{k-1}, t) = \gamma(t)$. Then by $b_\gamma([f'])$ is by definition represented by

$$D^k \rightarrow X, \quad y \mapsto \begin{cases} f'(2y), & \|y\| \leq \frac{1}{2}, \\ \gamma(2\|y\| - 1), & \|y\| \geq \frac{1}{2}. \end{cases}$$

This map is homotopic to g' relative S^{k-1} by virtue of the following homotopy.

$$D^k \times [0, 1] \rightarrow X, \quad y \mapsto \begin{cases} H'\left(\frac{2}{1+t}y, t\right), & \|y\| \leq \frac{1}{2}(1+t), \\ \gamma(2\|y\| - 1), & \|y\| \geq \frac{1}{2}(1+t). \end{cases}$$

□

Proof of Proposition A.2. Without loss of generality, W may be assumed to be connected. We first prove the absolute version ?? of the Proposition in three steps.

Step 1. We start by explaining how to kill a single homotopy class by surgery. Suppose that $w_0 \in W$ is an interior point. Let $\beta \in \pi_k(W, w_0)$ be a class represented by an embedding $f: (S^k, s_0) \rightarrow (W, w_0)$ for a specified basepoint $s_0 \in S^k$, which extends to an embedding $F: S^k \times D^{n-k} \rightarrow W$ into the interior of W . Let

$$\tilde{W} := W - \text{int}(\text{im}(F))$$

and define W' by the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ D^{k+1} \times S^{n-k-1} & \longrightarrow & W' \end{array}$$

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as discussed above. Since the inclusion $S^k \times S^{n-k-1} \rightarrow D^{k+1} \times S^{n-k-1}$ is a cofibration it follows that the above square is a homotopy pushout, hence the map $\tilde{W} \rightarrow W'$ has the same connectivity as $S^k \times S^{n-k-1} \rightarrow D^{k+1} \times S^{n-k-1}$, which is k -connected. Similarly, the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ S^k \times D^{n-k} & \longrightarrow & W \end{array}$$

implies that $\tilde{W} \rightarrow W$ is $n - k - 1$ -connected. Choose any point $d_0 \in \partial D^{n-k} = S^{n-k-1}$ and set $w'_0 := F(s_0, d_0)$. We now construct an epimorphism $\Psi_k: \pi_k(W, w_0) \rightarrow \pi_k(W', w'_0)$ mapping the class β to 0. Choose a path $\lambda: [0, 1] \rightarrow D^{n-k}$ from the center point $0 \in D^{n-k}$ to d_0 and let

$$\eta(t) := F(s_0, \lambda(t)).$$

Then η is a path from w_0 to w'_0 and we can consider the change of basepoint isomorphism

$$b_\eta: \pi_k(W, w_0) \rightarrow \pi_k(W, w'_0)$$

induced by η . From the preceding arguments it follows, that the map $\pi_i(\tilde{W}, w'_0) \rightarrow \pi_i(W, w'_0)$ induced by the inclusion is an isomorphism for $i \leq k$, since $2(k+1) \leq n$. Thus we can define maps

$$\Psi_i: \pi_i(W, w_0) \xrightarrow{b_\eta} \pi_i(W, w'_0) \xleftarrow{\cong} \pi_i(\tilde{W}, w'_0) \rightarrow \pi_i(W', w'_0).$$

By the above discussion Ψ_i is an isomorphism for $i \leq k-1$ and an epimorphism for $i = k$. We now show that Ψ_k maps the class β to 0. Define

$$\tilde{f}: (S^k, s_0) \rightarrow (W, w'_0), \quad y \mapsto F(y, d_0).$$

Then $\tilde{f}$ is freely homotopic to f via the homotopy $(y, t) \mapsto F(y, \lambda(t))$, so it follows from Lemma A.4 that $b_\eta(\beta)$ is represented by $\tilde{f}$. Note that the image of $\tilde{f}$ lies in $\tilde{W}$ and thus $\Psi_k(\beta)$ is represented by

$$(S^k, s_0) \rightarrow (W', w'_0), \quad y \mapsto \tilde{f}(y).$$

This map factors as the following composition of inclusions

$$S^k \rightarrow D^{k+1} \times \{d_0\} \rightarrow W',$$

which is clearly contractible. Therefore $\Psi_k(\beta) = 0$.

Step 2. Next we explain how to kill multiple homotopy classes at once. Let $\beta_i \in \pi_k(W, w_i)$ for interior points $w_i \in W$, $1 \leq i \leq l$. Again suppose that the β_i can be represented by embeddings $f_i: (S^k, s_0) \rightarrow (W, w_i)$, which extend to embeddings $F_i: S^k \times D^{n-k} \rightarrow W$ into the interior of W . Moreover assume these embeddings to be disjoint, that is $\text{im}(F_i) \cap \text{im}(F_j) = \emptyset$ for $i \neq j$. This allows us to do surgery along all F_i at once, i.e. we can define

$$\tilde{W} := W - \bigcup_{i=1, \dots, l} \text{int}(\text{im}(F_i))$$

and let W' be defined by the pushout

$$\begin{array}{ccc} \coprod_{i=1, \dots, l} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ \coprod_{i=1, \dots, l} D^{k+1} \times S^{n-k-1} & \longrightarrow & W' \end{array}$$

In order to make sense of the statement, that we can kill all homotopy classes β_i at once we would like to consider them as elements in the same group. To achieve this let $d_0 \in S^{n-k-1}$ and $\lambda: [0, 1] \rightarrow D^{n-k}$ be as in Step 1 and set

$$w'_i := F_i(s_0, d_0), \quad \eta_i(t) := F_i(s_0, \lambda(t)).$$

Moreover choose any point $v_0 \in \tilde{W}$ together with paths $\gamma'_i: [0, 1] \rightarrow \tilde{W}$ from $w'_i \rightarrow v_0$. Such paths exist, since W was assumed to be connected and removing (thickened) spheres of codimension at least 2 preserves connectedness. Let

$$\gamma_i := \gamma'_i * \eta_i: [0, 1] \rightarrow W$$

denote the composed path from w_i to v_0 and consider the following commutative diagram for $j \leq k$.

$$\begin{array}{ccccccc} \pi_j(W, v_0) & \xrightarrow{\text{id}} & \pi_j(W, v_0) & \xleftarrow{\cong} & \pi_j(\tilde{W}, v_0) & \longrightarrow & \pi_j(W', v_0) \\ b_{\gamma_i} \uparrow \cong & & b_{\gamma'_i} \uparrow \cong & & b_{\gamma'_i} \uparrow \cong & & b_{\gamma'_i} \uparrow \cong \\ \pi_j(W, w_i) & \xrightarrow{\cong} & \pi_j(W, w'_i) & \xleftarrow{\cong} & \pi_j(\tilde{W}, w'_i) & \longrightarrow & \pi_j(W', w'_i) \end{array}$$

Denote the map defined as the top horizontal composition by

$$\Phi_j: \pi_j(W, v_0) \rightarrow \pi_j(W', v_0).$$

The same arguments as in Step 1 show that Φ_j is an isomorphism for $j \leq k-1$ and an epimorphism for $j = k$. Moreover we have seen in Step 1, that for $j = k$, the class β_i is contained in the kernel of the bottom horizontal composition, hence $b_{\gamma_i}(\beta_i)$ is also contained in the kernel of Φ_k . Define

$$\phi := \Phi_1: \pi_1(W, v_0) \rightarrow \pi_1(W', v_0)$$

and note that all maps in the definition of Φ_k are compatible with the respective π_1 -actions, hence Φ_k is ϕ -equivariant.

Step 3. Finally we show how to kill multiple homotopy classes as formulated in ???. Let $g_i: (S^k, s_0) \rightarrow (W, w_0)$ be embeddings with trivial normal bundle representing the class α_i . By a general position argument we can find isotopies $H_i: S^k \times [0, 1] \rightarrow W$ from g_i to embeddings $f_i: S^k \rightarrow W - \{w_0\}$, such that $f_i(S^k) \cap f_j(S^k) = \emptyset$ for $i \neq j$. Moreover we may assume, that the image of f_i is contained in the interior of W . Since the normal bundle of g_i is trivial, the same holds for f_i . Therefore we can extend the f_i to embeddings

$$F_i: S^k \times D^{n-k} \rightarrow W - \{w_0\}$$

into the interior of W , such that $\text{im}(F_i) \cap \text{im}(F_j) = \emptyset$ for $i \neq j$. Now we are in the setting as in Step 2 for $w_i = f_i(s_0)$ and $\beta_i = [f_i] \in \pi_k(W, w_i)$. Define paths $\gamma_i(t) := H_i(s_0, 1-t)$ from w_i to w_0 . Then $b_{\gamma_i}(\beta_i) = \alpha_i$ by Lemma A.4. We would like to apply Step 2 to $v_0 = w_0$, however the paths γ_i are not of the same form as in Step 2 yet. Let us ignore this issue for a moment and see how Step 3 follows. Again let W' be the manifold obtained by surgery along $F_1, \dots, F_l$ as before. Step 2 now yields a homomorphism $\phi: \pi_1(W, w_0) \rightarrow \pi_1(W', w_0)$ together with a ϕ -equivariant epimorphism

$$\Phi_k: \pi_k(W, w_0) \rightarrow \pi_k(W', w_0)$$

containing the $b_{\gamma_i}(\beta_i) = \alpha_i$ in its kernel. Moreover, $\pi_j(W, w_0) \cong \pi_j(W', w_0)$ for $j \leq k-1$.

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We now construct a homotopy to bring γ_i into the desired form. Let $t_i := \sup\{t \in [0, 1] \mid \gamma_i([0, t]) \subseteq \text{im}(F_i)\}$ be the first time at which γ_i exits $\text{im}(F_i)$. For $t \in [0, t_i]$ we can then write $\gamma_i(t)$ in the form

$$\gamma_i(t) = F_i(\tilde{\sigma}_i(t), \tilde{\lambda}_i(t)).$$

for suitable paths $\tilde{\sigma}_i, \tilde{\lambda}_i$. Now $\tilde{\sigma}_i$ and $\tilde{\lambda}_i$ are homotopic to the reparametrized paths

$$\sigma_i(t) = \begin{cases} \tilde{\sigma}_i(0) = s_0, & t \in \left[0, \frac{t_i}{2}\right] \\ \tilde{\sigma}_i(2t - t_i), & t \in \left[\frac{t_i}{2}, t_i\right] \end{cases} \quad \text{and} \quad \lambda_i(t) = \begin{cases} \tilde{\lambda}_i(2t), & t \in \left[0, \frac{t_i}{2}\right] \\ \tilde{\lambda}_i(t_i), & t \in \left[\frac{t_i}{2}, t_i\right] \end{cases}$$

respectively. So after applying this homotopy we may replace $\tilde{\sigma}_i$ and $\tilde{\lambda}_i$ by σ_i and λ_i . For $t \in [0, \frac{t_i}{2}]$, $\gamma_i(t)$ can then be written as

$$\gamma_i(t) = F_i(s_0, \lambda_i(t)),$$

which is of the same form as η_i in Step 2. It remains to bring $\gamma'_i := \gamma_i|_{[\frac{t_i}{2}, 1]}$ into the desired form, meaning the image of γ'_i should be completely contained within $\tilde{W}$. We can achieve this as follows. Applying a general position argument we can assume, that γ'_i does not intersect any of the embedded spheres $f_1(S^k), \dots, f_l(S^k)$. Consider the deformation retraction

$$r: W - \bigcup_{j=1, \dots, l} f_j(S^k) \rightarrow \tilde{W}$$

induced by the deformation retractions $S^k \times (D^{n-k} - 0) \rightarrow S^k \times S^{n-k-1}$. Then $r \circ \gamma'_i$ is a path in $\tilde{W}$, which is homotopic to γ'_i relative to the endpoints. So in conclusion we may indeed assume, that γ_i is of the form as in Step 2.

Step 4. We can now deduce the relative version ?? of the proposition using ?? applied to the given classes $\alpha_1, \dots, \alpha_l \in \pi_k(W, x_0)$. Using these classes construct $\tilde{W}$ and W' as before. For $i \leq k$ consider the following diagram, where all maps are induced by inclusions and the rows are exact.

$$\begin{array}{ccccccccc} \pi_i(M, x_0) & \longrightarrow & \pi_i(W, x_0) & \longrightarrow & \pi_i(W, M, x_0) & \longrightarrow & \pi_{i-1}(M, x_0) & \longrightarrow & \pi_{i-1}(W, x_0) \\ \uparrow \text{id} & & \uparrow \cong & & \uparrow & & \uparrow \text{id} & & \uparrow \cong \\ \pi_i(M, x_0) & \longrightarrow & \pi_i(\tilde{W}, x_0) & \longrightarrow & \pi_i(\tilde{W}, M, x_0) & \longrightarrow & \pi_{i-1}(M, x_0) & \longrightarrow & \pi_{i-1}(\tilde{W}, x_0) \\ \downarrow \text{id} & & \downarrow & & \downarrow & & \downarrow \text{id} & & \downarrow \cong \\ \pi_i(M, x_0) & \longrightarrow & \pi_i(W', x_0) & \longrightarrow & \pi_i(W', M, x_0) & \longrightarrow & \pi_{i-1}(M, x_0) & \longrightarrow & \pi_{i-1}(W', x_0) \end{array}$$

It follows from the Five Lemma that

$$\pi_i(\tilde{W}, M, x_0) \rightarrow \pi_i(W, M, x_0)$$

is an isomorphism and

$$\pi_i(\tilde{W}, M, x_0) \rightarrow \pi_i(W', M, x_0)$$

is an isomorphism for $i \leq k-1$. This shows $\pi_i(W, M, x_0) \cong \pi_i(W', M, x_0)$ for $i \leq k-1$. If $i = k$ the map $\pi_k(\tilde{W}, x_0) \rightarrow \pi_k(W', x_0)$ is surjective and a diagram chase shows that

$$\pi_k(\tilde{W}, M, x_0) \rightarrow \pi_k(W', M, x_0)$$

is surjective as well. Thus we can define the epimorphism

$$\Phi_{\text{rel}}: \pi_k(W, M, x_0) \xleftarrow{\cong} \pi_k(\tilde{W}, M, x_0) \rightarrow \pi_k(W', M, x_0).$$

We have already seen that $\alpha_1, \dots, \alpha_l$ are contained in the kernel of the map

$$\pi_k(W, x_0) \xleftarrow{\cong} \pi_k(\tilde{W}, x_0) \rightarrow \pi_k(W', x_0).$$

So commutativity of the above diagram implies

$$\Phi_{\text{rel}}(\alpha_i^{\text{rel}}) = 0.$$

Note that all maps in the definition of Φ_{rel} are $\pi_1(M, x_0)$ -equivariant, such that the same holds for Φ_{rel} . $\square$

A.2. A Result about Homotopy Groups of CW-Pairs

In this section we give a detailed proof of the following proposition, which was used in the proof of Theorem 2.11 and Theorem 2.12.

Proposition A.5. *Let (X, A) be a CW pair with X connected. Suppose that (X, A) is $(n-1)$ -connected for $n \geq 2$, that the inclusion $A \rightarrow X$ induces an isomorphism on π_1 , and that (X, A) contains only finitely many n -cells. Then $\pi_n(X, A)$ is finitely generated as a $\mathbb{Z}[\pi_1(A)]$ -module.*

The proof relies on the following purely algebraic result, which is an adaptation of Lemma 1.12 in [Pedersen].

Lemma A.6. *Let P_* be a positive chain complex of projective R -modules, which is homologically $(n-1)$ -connected, that is $P_i = 0$ for $i < 0$ and $H_k(P_*) = 0$ for $k \leq n-1$. Suppose that P_n is finitely generated. Then $H_n(P_*)$ is also finitely generated.*

Proof. Denote by $\partial_k: P_k \rightarrow P_{k-1}$ the k -th boundary map. We show inductively that $P_k \cong \ker(\partial_k) \oplus \ker(\partial_{k-1})$ for $0 \leq k \leq n$. The base case $k = 0$ is trivially true, so suppose the claim holds for some $k < n$. Then $\ker(\partial_k)$ is projective as a direct summand of a projective module and $H_k(P_*) = 0$ is equivalent to the fact that $\partial_{k+1}: P_{k+1} \rightarrow \ker(\partial_k)$ is surjective. Hence $P_{k+1} \cong \ker(\partial_{k+1}) \oplus \ker(\partial_k)$. So in particular

$$P_n \cong \ker(\partial_n) \oplus \ker(\partial_{n-1}),$$

such that $\ker(\partial_n)$ is finitely generated as a direct summand of a finitely generated module. Consequently the quotient $H_n(P_*)$ is also finitely generated. $\square$

Moreover we need the following fact concerning the homology of relative CW-complexes.

Lemma A.7. *Let Y, B be topological spaces, where Y is obtained from B by attaching finitely many n -cells. Let $p: \hat{Y} \rightarrow Y$ be some cover, such that the group $\text{Deck}(p)$ of deck transformations of p acts transitively on the fibers. Set $\hat{B} := p^{-1}(B)$. Then $H_n(\hat{Y}, \hat{B})$ is finitely generated as a $\mathbb{Z}[\text{Deck}(p)]$ -module, where the action of an element $\phi \in \text{Deck}(p)$ is given by $\phi_*: H_n(\hat{Y}, \hat{B}) \rightarrow H_n(\hat{Y}, \hat{B})$.*

Proof. Let us recall how the relative CW-structure of (Y, B) lifts to a relative CW-structure of $(\hat{Y}, \hat{B})$. Let I be the index set for the n -cells of (Y, B) with characteristic maps $Q_i: D^n \rightarrow Y$, $i \in I$. Choose some fixed basepoint $e_0 \in D^n$. Then for each $y \in \hat{Y}$ with $p(y) = Q_i(e_0)$, there exists a unique lift $Q_i^y: D^n \rightarrow \hat{Y}$ of Q_i along p , such that $Q_i^y(e_0) = y$. These Q_i^y indexed over the set $\Lambda := \{(y, i) \in \hat{Y} \times I \mid p(y) = Q_i(e_0)\}$ are then the characteristic maps for the n -cells of $(\hat{Y}, \hat{B})$. The corresponding attaching maps are the restrictions $q_i^y := Q_i^y|_{S^{n-1}}: S^{n-1} \rightarrow \hat{B}$. See [FP90, Theorem 1.1.6] for further details. From excision in the pushout

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$$\begin{array}{ccc} \coprod_{\Lambda} S^{n-1} & \xrightarrow{(q_i^y)} & \tilde{B} \\ \downarrow & & \downarrow \\ \coprod_{\Lambda} D^n & \xrightarrow{(Q_i^y)} & \hat{Y} \end{array}$$

we obtain an isomorphism

$$\mathbf{Q}: \bigoplus_{\Lambda} H_n(D^n, S^{n-1}) \rightarrow H_n(\hat{Y}, \hat{B})$$

An element $\phi \in \text{Deck}(p)$ acts on $\bigoplus_{\Lambda} H_n(D^n, S^{n-1})$ by mapping the summand with index (y, i) to the summand with index $(\phi(y), i)$. We claim that $\mathbf{Q}$ is equivariant with respect to this action and the previously defined $\text{Deck}(p)$ -action on $H_n(\hat{Y}, \hat{B})$. This follows from commutativity of the following diagram, where again the action of an element $\phi \in \text{Deck}(p)$ on $\coprod_{\Lambda} D^n$ and $\coprod_{\Lambda} S^{n-1}$ is given by permuting indices as above:

$$\begin{array}{ccccc} & & \coprod_{\Lambda} S^{n-1} & \xrightarrow{(q_i^y)} & \tilde{B} \\ & \nearrow \phi & \downarrow (q_i^y) & & \downarrow \phi \\ \coprod_{\Lambda} S^{n-1} & \xrightarrow{\quad} & \tilde{B} & & \tilde{Y} \\ & \downarrow & \downarrow (Q_i^y) & \searrow \phi & \\ & \coprod_{\Lambda} D^n & \xrightarrow{\quad} & \tilde{Y} & \\ \downarrow & \nearrow \phi & \downarrow (Q_i^y) & & \\ \coprod_{\Lambda} D^n & \xrightarrow{\quad} & \tilde{Y} & & \end{array}$$

Therefore $\mathbf{Q}$ is an isomorphism of $\mathbb{Z}[\text{Deck}(p)]$ -modules. But since $\text{Deck}(p)$ acts transitively on the fibers of p and we assumed the index set I to be finite, $\bigoplus_{\Lambda} H_n(D^n, S^{n-1})$ is a finitely generated $\mathbb{Z}[\text{Deck}(p)]$ -module (for each $i \in I$ choose one $(y, i) \in \Lambda$ and a generator of the summand $H_n(D^n, S^{n-1})$ corresponding to this index (y, i)). $\square$

Proof of Proposition A.5. Let $p: \tilde{X} \rightarrow X$ be the universal cover. Since the inclusion $A \rightarrow X$ induces an isomorphism on π_1 the universal cover of A is given by $\tilde{A} \cong p^{-1}(A)$. Choosing basepoints $x_0 \in A$ and $\tilde{x}_0 \in \tilde{A}$ with $p(\tilde{x}_0) = x_0$, we obtain canonical isomorphisms

$$\pi_1(A, x_0) \cong \pi_1(X, x_0) \cong \text{Deck}(p),$$

where $\text{Deck}(p)$ is the group of deck transformations of p . By virtue of these isomorphisms we denote any of the three groups simply by π . Consider the following actions of π . On $\pi_n(X, A, x_0)$ we have the natural $\pi_1(A, x_0)$ action. On $H_n(\tilde{X}, \tilde{A})$ consider the action

$$\text{Deck}(p) \ni \phi \mapsto \phi_*: H_n(\tilde{X}, \tilde{A}) \rightarrow H_n(\tilde{X}, \tilde{A})$$

and on $\pi_n(\tilde{X}, \tilde{A}, \tilde{x}_0)$ the action

$$\text{Deck}(p) \ni \phi \mapsto \phi_*: \pi_n(\tilde{X}, \tilde{A}, \tilde{x}_0) \rightarrow \pi_n(\tilde{X}, \tilde{A}, \phi(\tilde{x}_0)) \cong \pi_n(\tilde{X}, \tilde{A}, \tilde{x}_0),$$

where the second isomorphism is the canonical change of basepoint isomorphism ($\tilde{A}$ is simply connected). Consider the isomorphisms

$$p_*: \pi_n(\tilde{X}, \tilde{A}, \tilde{x}_0) \rightarrow \pi_n(X, A, x_0)$$

and

$$h: \pi_n(\tilde{X}, \tilde{A}, \tilde{x}_0) \rightarrow H_n(\tilde{X}, \tilde{A}),$$

where h is the Hurewicz map. Note that h is an isomorphism by the Hurewicz Theorem since $\pi_k(\tilde{X}, \tilde{A}, \tilde{x}_0) \cong \pi_k(X, A, x_0) = 0$ for $2 \leq k \leq n-1$ and $\tilde{X}, \tilde{A}$ are simply connected. These isomorphisms are now equivariant with respect to the π -actions defined above. For p_* this is a direct consequence of the definitions and for h this follows from naturality and the fact that $h([f])$ only depends on the free homotopy class of $f: (D^n, S^{n-1}) \rightarrow (\tilde{X}, \tilde{A})$, which is invariant under the change of basepoint isomorphism. So in other words, p_* and h are isomorphisms of $\mathbb{Z}[\pi]$ -modules (note that even for $n=2$ all groups are abelian). Therefore it suffices to show that $H_n(\tilde{X}, \tilde{A})$ is a finitely generated $\mathbb{Z}[\pi]$ -module. Let $C_*^{\text{cell}}(\tilde{X}, \tilde{A})$ be the cellular chain complex of $(\tilde{X}, \tilde{A})$ with respect to the relative CW-structure induced by (X, A) . If X_k denotes the k -skeleton of (X, A) the k -skeleton of $(\tilde{X}, \tilde{A})$ is given by $\tilde{X}_k := p^{-1}(X_k)$, compare the proof of Lemma A.7 or [FP90, Proposition 2.3.9]. By assumption, X_n is obtained from X_{n-1} by attaching finitely many n -cells. Therefore we can apply Lemma A.7 to the cover $p: \tilde{X}_n \rightarrow X_n$ to conclude that

$$C_n^{\text{cell}}(\tilde{X}, \tilde{A}) = H_n(\tilde{X}_n, \tilde{X}_{n-1}),$$

is a finitely generated $\mathbb{Z}[\pi]$ -module. Moreover, it follows from the exact sequence

$$H_1(\tilde{X}) \rightarrow H_1(\tilde{X}, \tilde{A}) \rightarrow H_0(\tilde{A}) \rightarrow H_0(\tilde{X}) \rightarrow H_0(\tilde{X}, \tilde{A})$$

that $H_0(\tilde{X}, \tilde{A}) = H_1(\tilde{X}, \tilde{A}) = 0$, since both $\tilde{X}$ and $\tilde{A}$ are connected and $\tilde{X}$ is simply connected. Furthermore,

$$H_k(\tilde{X}, \tilde{A}) \cong \pi_k(\tilde{X}, \tilde{A}, \tilde{x}_0) \cong \pi_k(X, A, x_0) = 0$$

for $2 \leq k \leq n-1$. Note that, by naturality, the boundary maps of the chain complex $C_*^{\text{cell}}(\tilde{X}, \tilde{A})$ are also $\mathbb{Z}[\pi]$ linear. Hence $C_*^{\text{cell}}(\tilde{X}, \tilde{A})$ satisfies the conditions of Lemma A.6 and we conclude that $H_n(\tilde{X}, \tilde{A})$ is indeed a finitely generated $\mathbb{Z}[\pi]$ -module. $\square$

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